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Chapter 2

THE PHYSICAL LAYER

Contents



1. Introduction
2. **The Physical Layer**
3. The Data Link Layer
4. The Medium Access Control Sub-Layer
5. The Network Layer
6. The Transport Layer
7. The Application Layer
8. Network Security

DLL: Data Link Layer

- LLC: Logical Link Control
- MAC: Medium Access Control

Chapter 7 Application

Chapter 6 Transport

Chapter 5 Network

Chapter 3 LLC

Chapter 4 MAC

Data Link

Chapter 2 Physical

Contents



1. The Theoretical Basis for Data Communication
2. Guided Transmission Media
3. Wireless Transmission
4. Communication Satellites
5. Digital Modulation and Multiplexing
6. The Public Switched Telephone Network
7. The Mobile Telephone System
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2.1 The Theoretical Basis for Data Communication



- 2.1.1 Fourier Analysis
- 2.1.2 Bandwidth-Limited Signals
- 2.1.3 The Maximum Data Rate of a Channel

2.1.1 Fourier Analysis



➤ Fourier series

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(n \cdot 2\pi f t) + \sum_{n=1}^{\infty} b_n \cos(n \cdot 2\pi f t)$$

$$\left\{ \begin{array}{l} a_n = \frac{2}{T} \int_0^T g(t) \sin(n \cdot 2\pi f t) dt \\ b_n = \frac{2}{T} \int_0^T g(t) \cos(n \cdot 2\pi f t) dt \\ c = \frac{2}{T} \int_0^T g(t) dt \end{array} \right.$$

2.1.2 Bandwidth-Limited Signals

- Bandwidth, 带宽
 - ✦ The width of the frequency range transmitted without being strongly attenuated.
- Baseband signals, 基带信号
 - ✦ Signals that run from 0 up to a maximum frequency
- Passband signals, 频带信号, 通带信号
 - ✦ Signals that are shifted to occupy a higher range of frequencies

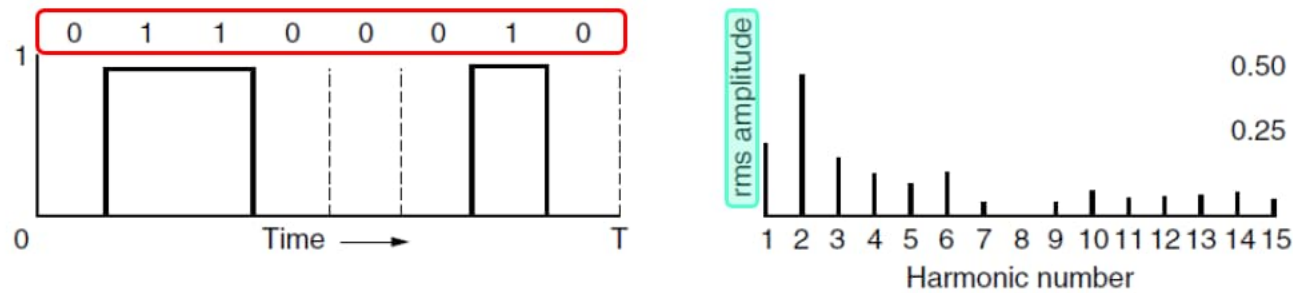


Figure 2-1. (a) A binary signal and its root-mean-square Fourier amplitudes.

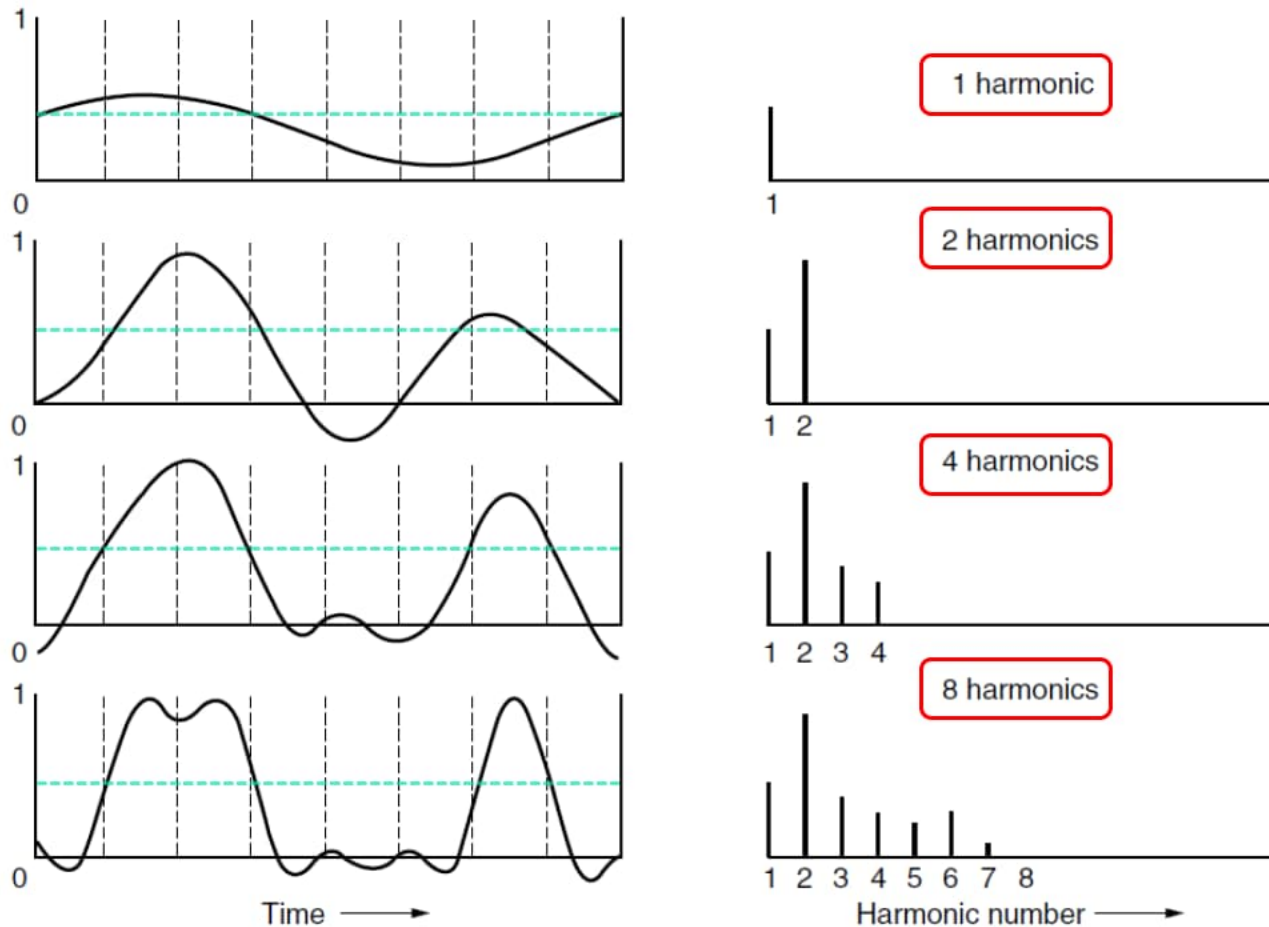


Figure 2-1. (b)–(e)
Successive
approximations to
the original signal.

rms: root-mean-square,
方均根值

Sending 8 bits



data rate (bps) b	Time (msec) 8/b	First harmonic (Hz) b/8	# Harmonics sent 3000/(b/8)
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

Figure 2-2. Relation between data rate and harmonics for an ordinary telephone line, with cutoff frequency just above 3000 Hz.

2.1.3 The Maximum Data Rate of a Channel



➤ Nyquist's theorem

- ✦ for an arbitrary signal has been run through a low-pass filter of **bandwidth B** within **noiseless** channel
- ✦ If the signal consists of **V discrete levels**
 - maximum data rate $R = 2B \cdot \log_2 V$ (bps)

➤ Shannon's theorem

- ✦ a **noisy** channel whose **bandwidth** is **B Hz** and whose **signal-to-noise ratio** is **S/N decibels (dB)**,
- ✦ maximum data rate or capacity $C = B \cdot \log_2(1 + \frac{S}{N})$ (bps)

➤ The final limit depends on the **minimum value** of the two

dB: deci Bel (分贝)

2.2 Guided Transmission Media

- 2.2.1 Magnetic Media
- 2.2.2 Twisted Pairs
- 2.2.3 Coaxial Cable
- 2.2.4 Power Lines
- 2.2.5 Fiber Optics

Guided Transmission Media

引导性传输介质

2.2.1 Magnetic Media



- One of the most common ways
 - ✦ to **transport data** from one computer to another
 - is to write them onto **magnetic tape** or removable media (e.g., recordable DVDs),
 - **physically transport** the tape or disks to the destination machine, and **read** them back in **again**.

- *Never underestimate the bandwidth of a station wagon full of tapes hurtling down the highway.*

2.2.2 Twisted Pairs

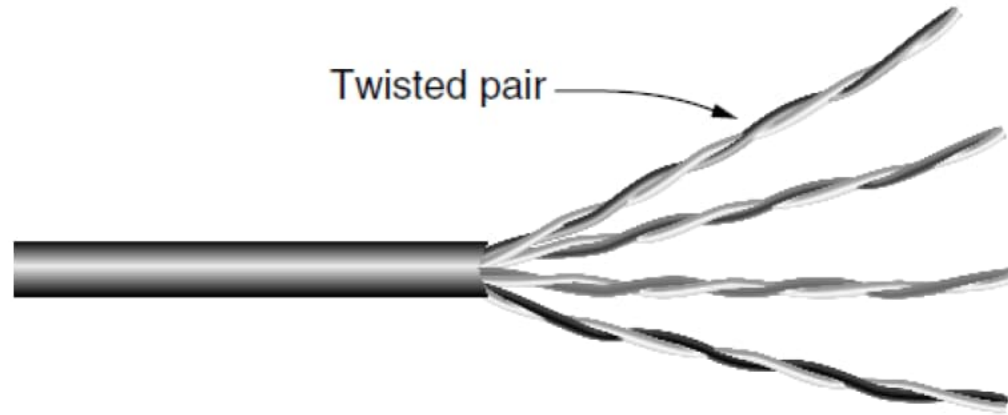


Figure 2-3. Category 5 UTP cable with four twisted pairs.

twisted pair: 双绞线

Terminologies



- Full-duplex links (全双工链路)
 - ✦ Links that can be used in **both** directions **at the same time**
 - like a two-lane road
- Half-duplex links (半双工链路)
 - ✦ links that can be used in **either** direction, but **only one way** at a time
 - like a single-track railroad line
- Simplex links (单工链路)
 - ✦ links that allow traffic in **only one** direction
 - like a one-way street
- UTP: Unshielded Twisted Pair (非屏蔽双绞线)
- STP: Shielded Twisted Pair (屏蔽双绞线)

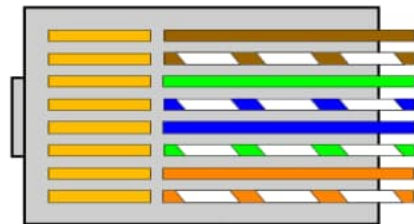
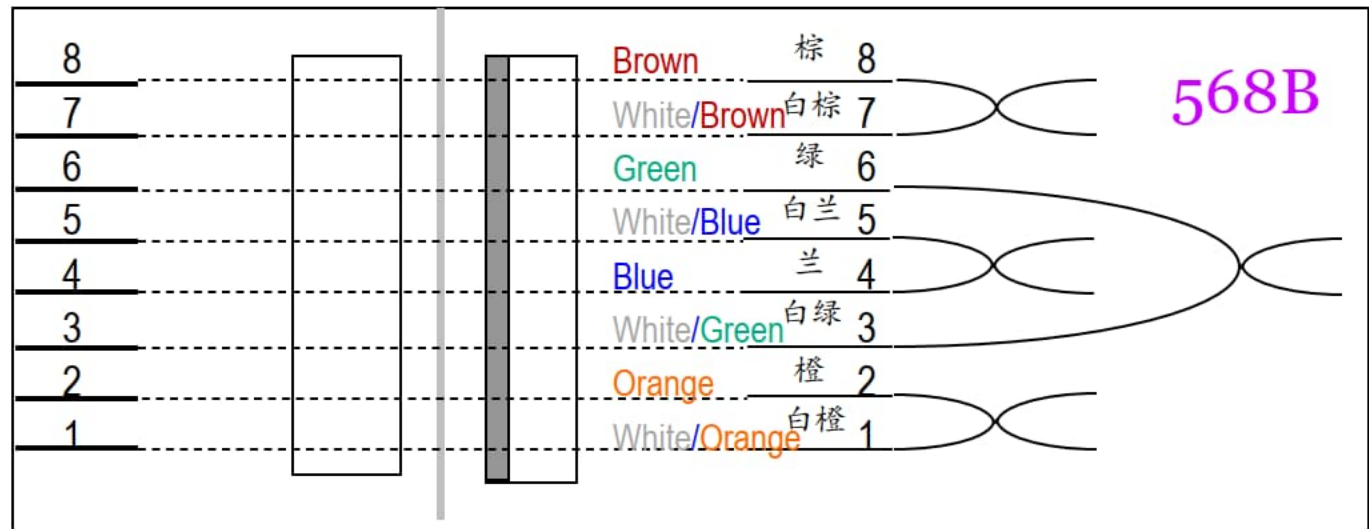
Categories of Unshielded Twisted Pair



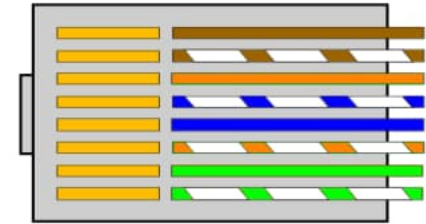
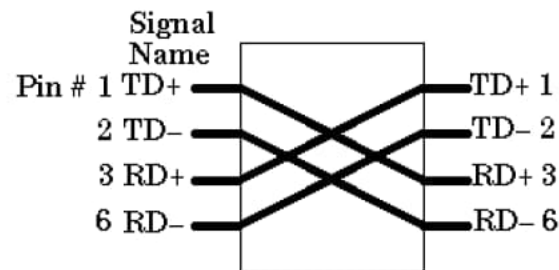
- The EIA/TIA (Electronic Industry Association/Telecommunication Industry Association) has established standards of UTP.

★ Type	Use
★ Category 1	Voice Only (Telephone Wire)
★ Category 2	Data to 4 Mbps (LocalTalk)
★ Category 3	Data to 10 Mbps (Ethernet)
★ Category 4	Data to 20 Mbps (16 Mbps Token Ring)
★ Category 5	Data to 100 Mbps (Fast Ethernet)
★ Category 5e	Data to 100 Mbps (Fast Ethernet)
★ Category 6	Data to 250 MHz (Gigabit Ethernet) (4D-PAM5 line coding)
★ Category 7	Data to 600 MHz

568B/568A standard



568B



568A

2.2.3 Coaxial Cable

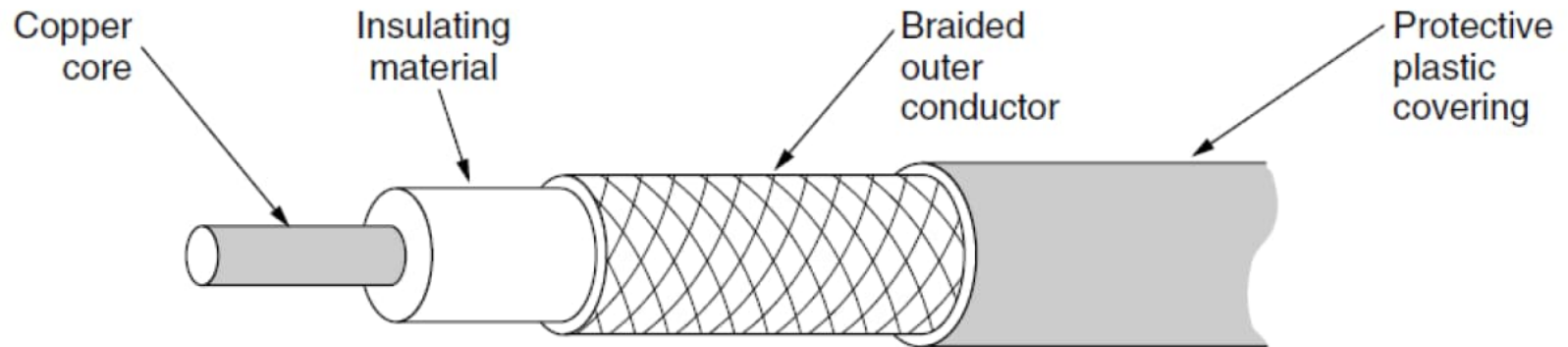
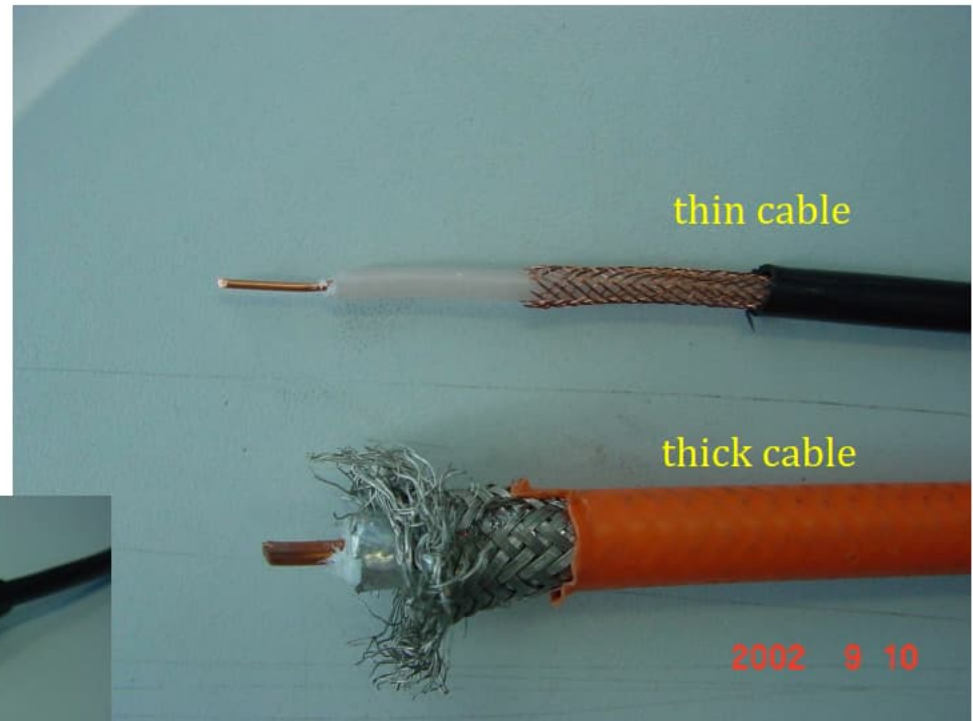


Figure 2-4. A coaxial cable.

braid 英 [breɪd] 美 [breɪd] n. 辫子; 编织物; 穗带; 发辫; vt. 编织;

Thick and thin cable



2.2.4 Power Lines

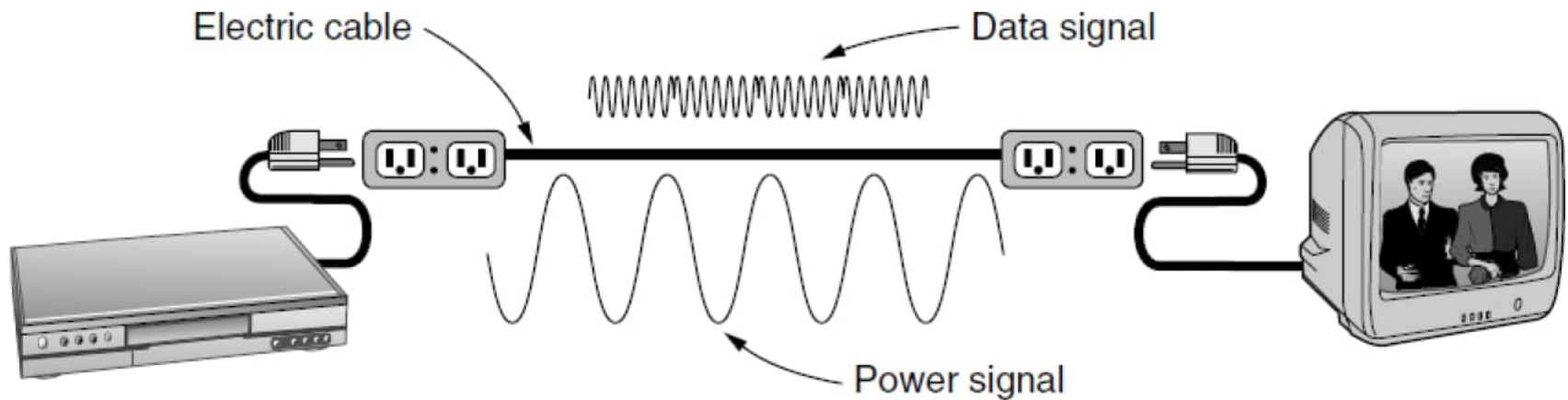
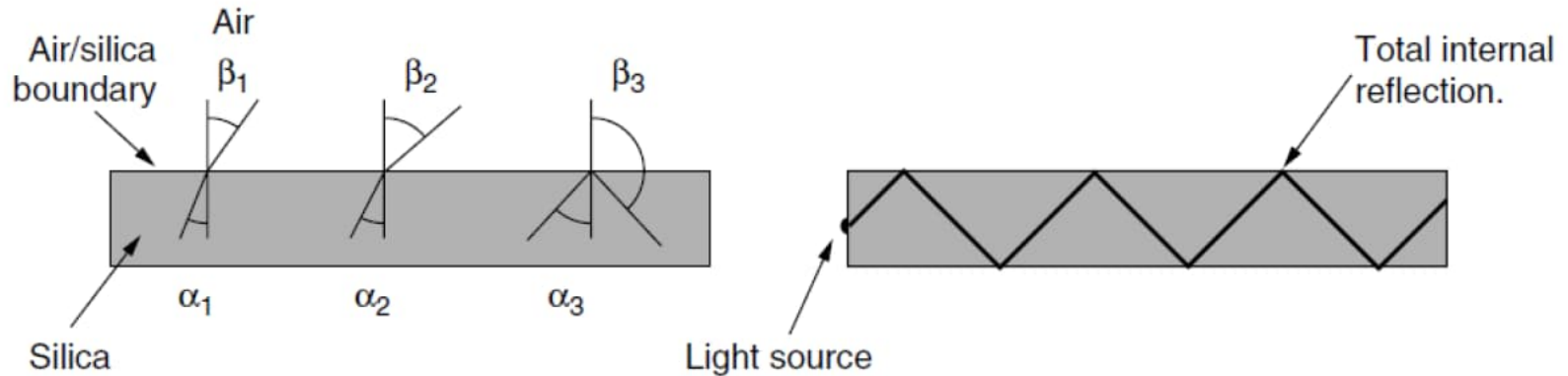


Figure 2-5. A network that uses household electrical wiring.

2.2.5 Fiber Optics



(a) Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles

(b) Light trapped by total internal reflection.

Figure 2-6.

silica ['sɪlkə] n. 二氧化硅

Transmission of Light Through Fiber

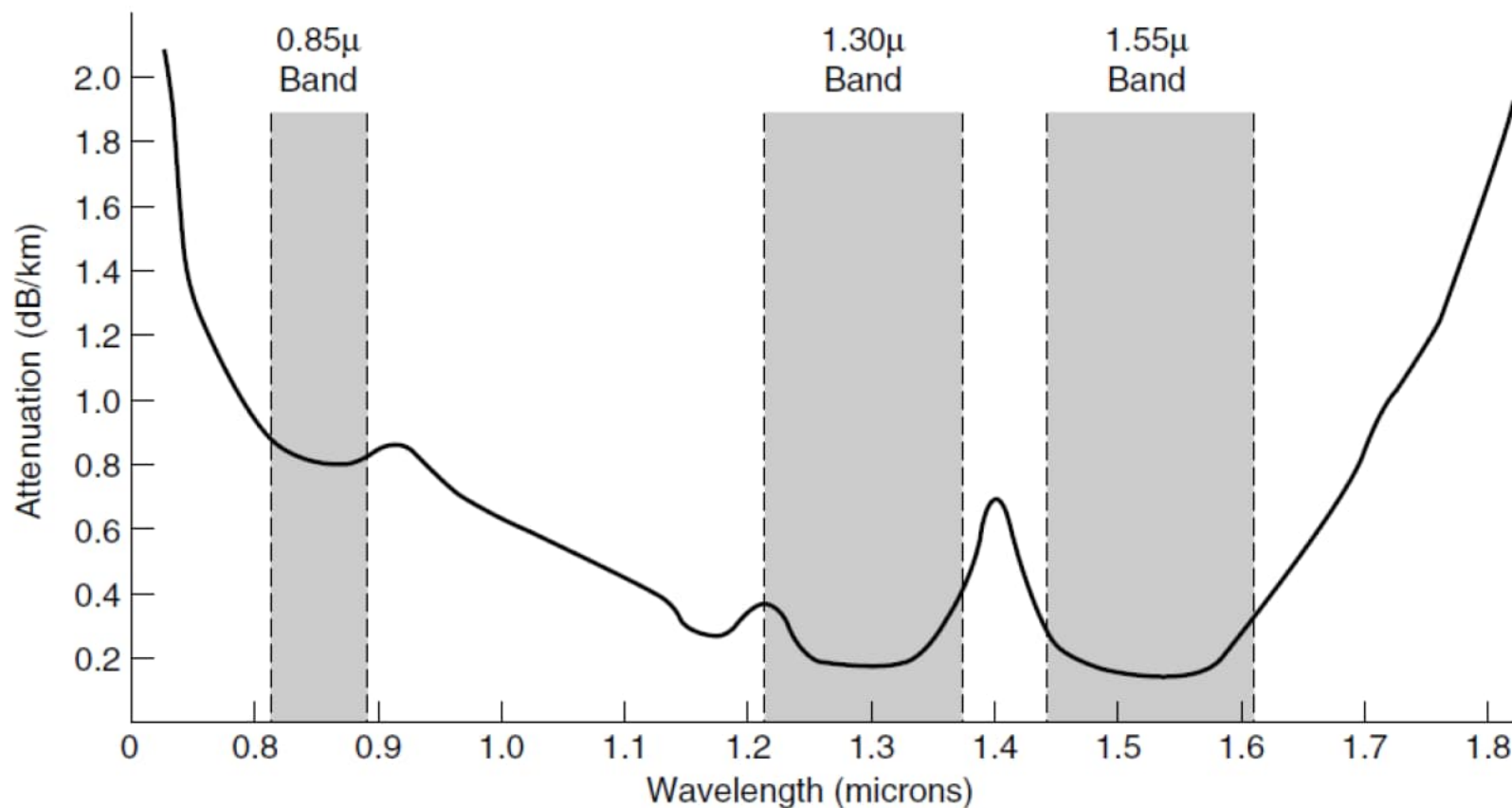
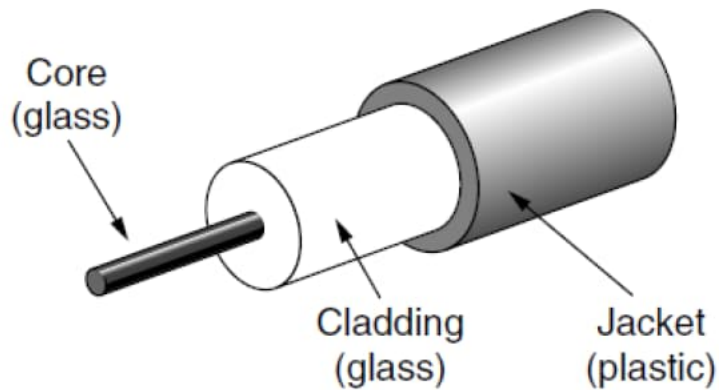
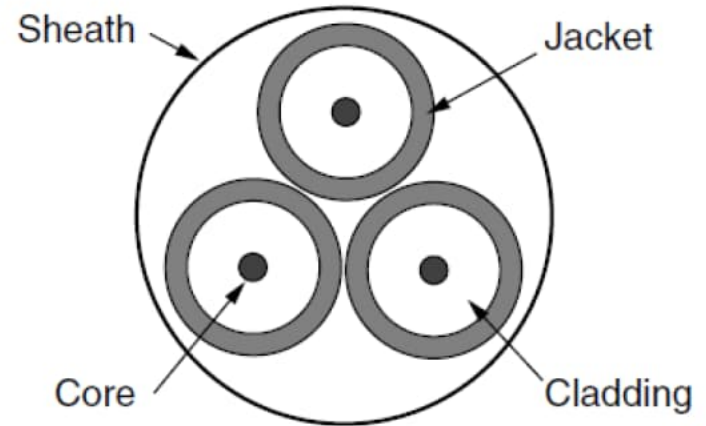


Figure 2-7. Attenuation of light through fiber in the infrared region.

Fiber Cables



(a) Side view of a single fiber.



(b) End view of a sheath with three fibers.

Figure 2-8.

clad [klæd]

adj. 覆盖的; v. 穿 (衣)

jacket ['dʒækɪt]

n. 夹克; 护封; 文件套

sheath 英 [ʃi:θ] 美 [ʃiθ]

n. 鞘; 护套; v. 包, 盖

Optical fiber modes

➤ Multi-mode fibers, 多模光纤

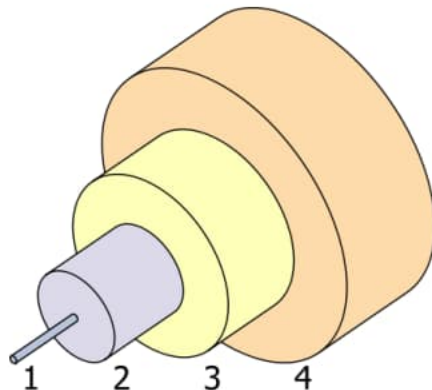
- ★ Multi-mode fibers are described by their core and cladding diameters.
 - typical 62.5/125 μm , 50/125 μm



➤ Single-mode fiber (SMF), 单模光纤

- ★ A typical single-mode optical fiber has a core diameter between 8 and 10.5 μm and a cladding diameter of 125 μm .

- ★ The structure of a typical single-mode fiber.



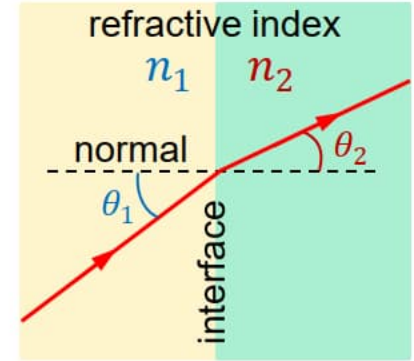
1. Core 8 μm diameter
2. Cladding 125 μm dia.
3. Buffer 250 μm dia.
4. Jacket 400 μm dia.

Refractive Index Profile



➤ Refractive index, 折射率

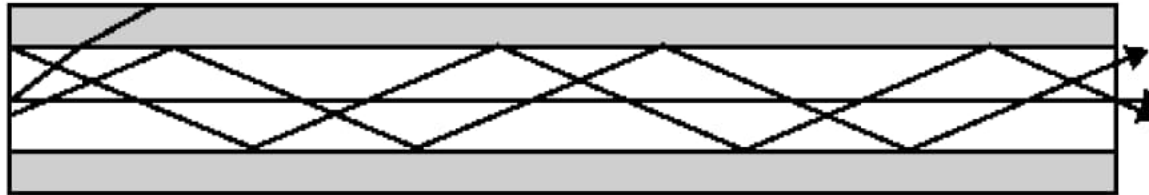
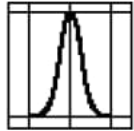
- ✦ describes how light propagates through the medium.
- ✦ It is defined as: $n = \frac{c}{v}$



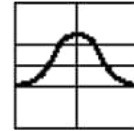
➤ Refractive Index Profile, 折射率分布图

- ✦ Step-index fiber, 阶跃折射率
 - a uniform refractive index within the core and a sharp decrease in refractive index at the core-cladding interface.
- ✦ Graded-index fiber, 渐变折射率
 - whose core has a refractive index that decreases with increasing radial distance from the optical axis of the fiber.

Input pulse

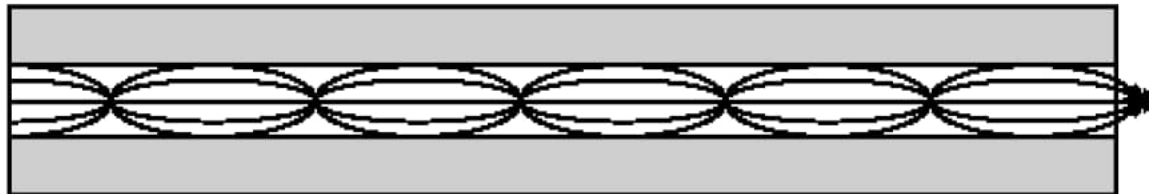


Output pulse

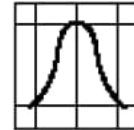


(a) Step-index multimode

Input pulse



Output pulse

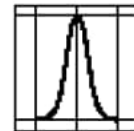


(b) Graded-index multimode

Input pulse



Output pulse



(c) Single mode

Light sources



Item	LED	Semiconductor Laser
Data rate	Low	High
Fiber type	Multi-mode	Multi-mode or single-mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

Figure 2-9. A comparison of **semiconductor** diodes and **LEDs** as light sources.

Comparison of Fiber Optics and Copper Wire



➤ Advantages

- ✦ much **higher bandwidths** than copper
- ✦ **low attenuation**
 - repeaters are needed only about every **50 km** on long lines, versus about every **5 km** for copper.
- ✦ **not being affected** by
 - power surges, electromagnetic interference, or power failures, ...
- ✦ **thin and lightweight**
- ✦ **do not leak light** and are **difficult to tap**

power surges 电涌; 浪涌;

➤ Disadvantages

- ✦ can **be damaged** easily by **being bent** too much
- ✦ **unidirectional**
- ✦ fiber **interfaces cost more** than electrical interfaces

2.3 Wireless Transmission

- 2.3.1 The Electromagnetic Spectrum
- 2.3.2 Radio Transmission
- 2.3.3 Microwave Transmission
- 2.3.4 Infrared Transmission
- 2.3.5 Light Transmission

2.3.1 The Electromagnetic Spectrum

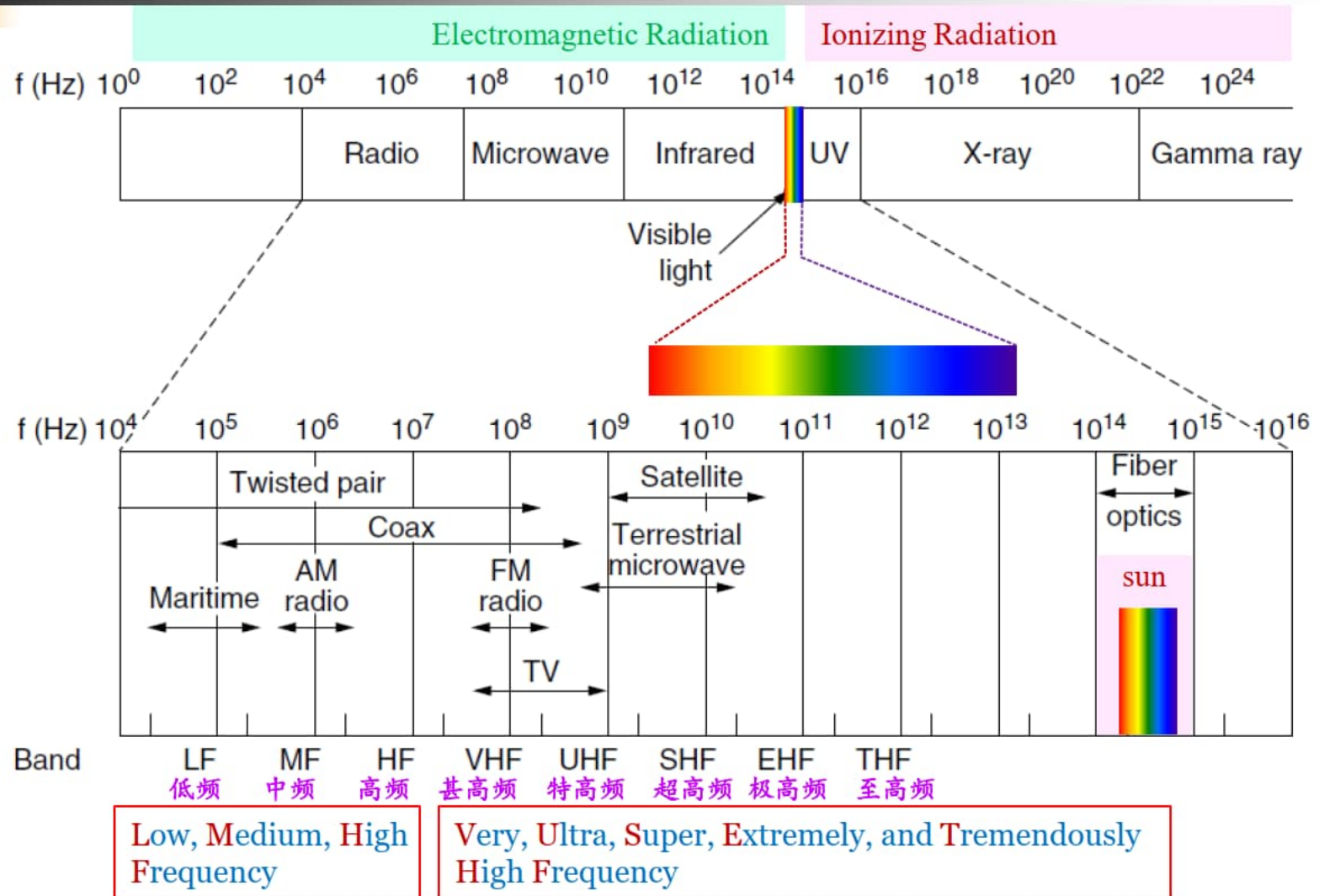
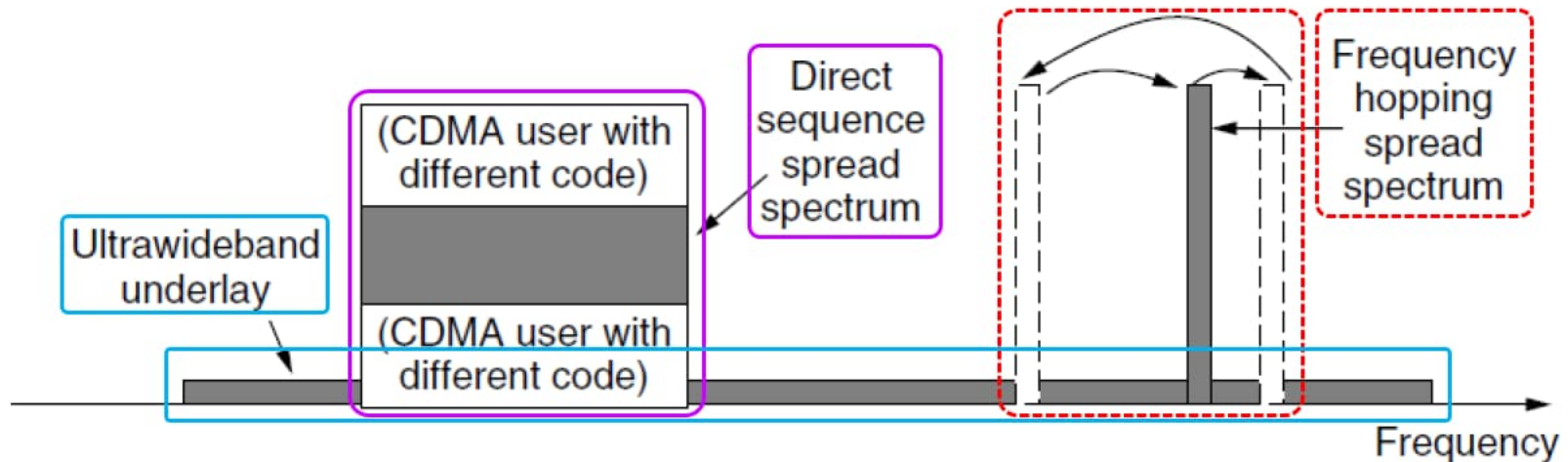


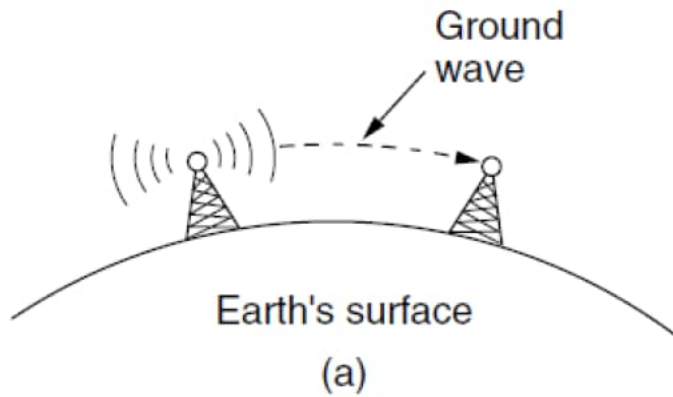
Figure 2-10. The electromagnetic spectrum and its uses for communication.

Narrow & Wide frequency band

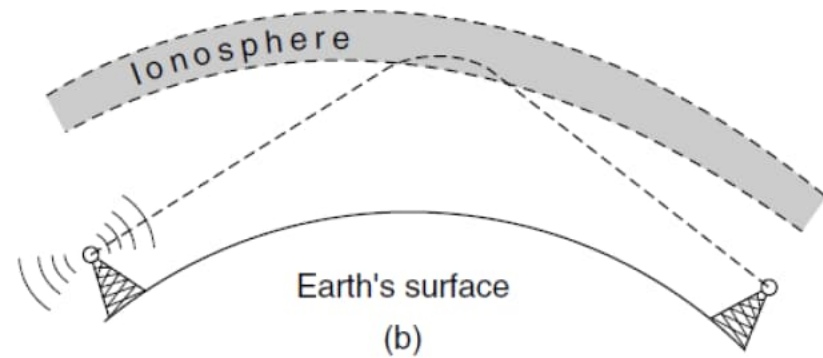
- Narrow frequency band 窄带
 - ✦ concentrate to spectrum efficiency, data rates
- Wide frequency band 宽带
 - ✦ frequency hopping spread spectrum 跳频扩频
 - ✦ direct sequence spread spectrum 直序扩频
 - CDMA: Code Division Multiple Access 码分多址
- UWB (Ultra-WideBand) 超宽带



2.3.2 Radio Transmission



(a) In the VLF, LF, and MF bands, radio waves follow the curvature



(b) In the HF band, radio waves bounce off the ionosphere.

Figure 2-12.

2.3.3 Microwave Transmission

- Above 100 MHz
 - ✦ the waves travel in **nearly straight lines** and can therefore be **narrowly focused**.
 - ✦ **Repeaters** are needed **periodically**, The higher the towers are, the farther apart they can be.
 - ✦ **absorption** by water at about **4 GHz**
- Key advantages over fiber
 - ✦ **no right of way** is needed
 - ✦ relatively **inexpensive**
- widely used for
 - ✦ long-distance telephone, mobile phones, television,
 - ✦ severe shortage of spectrum

The Politics of the Electromagnetic Spectrum



➤ for unlicensed usage

- ✦ ISM (Industrial, Scientific, Medical) bands
- ✦ U-NII (Unlicensed National Information Infrastructure) bands

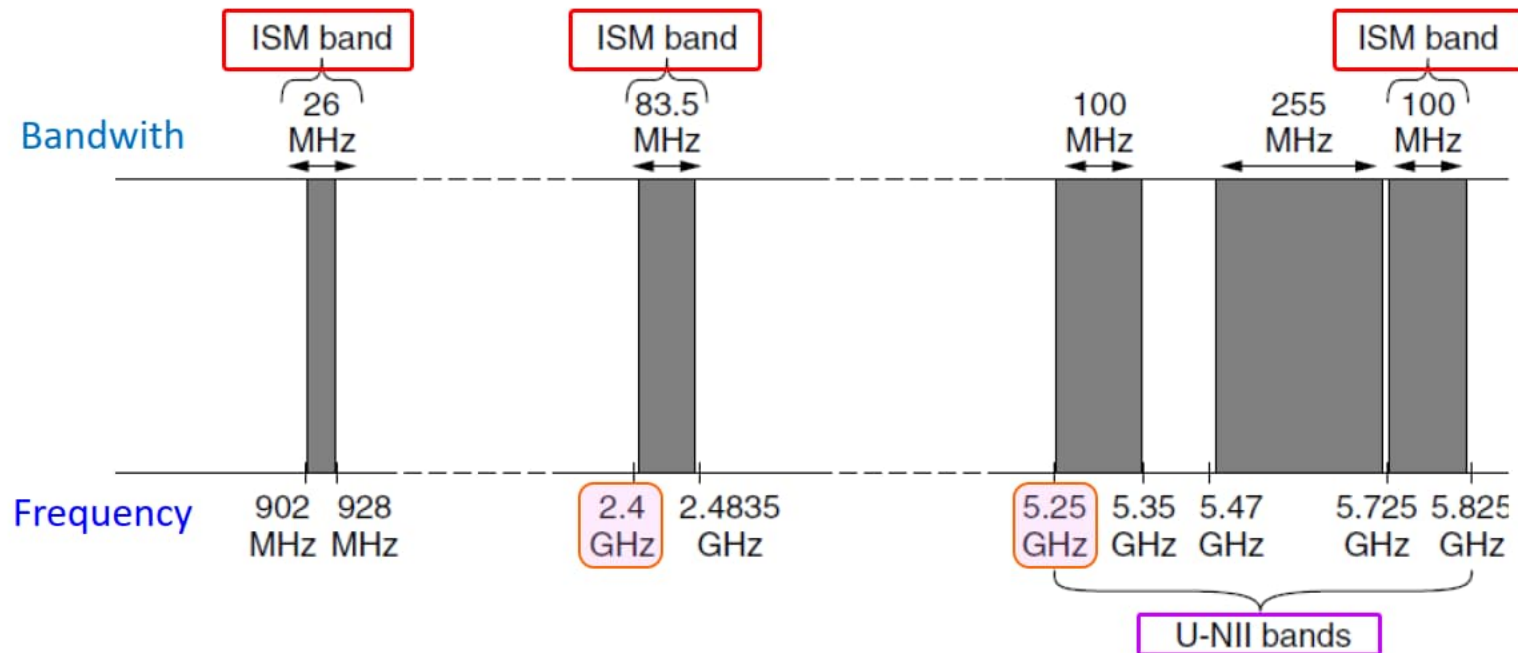


Figure 2-13. ISM and U-NII bands used in the United States by wireless devices.

2.3.4 Infrared Transmission

- short-range communication
 - ✦ remote controls used for televisions, VCRs, and stereos
- Advantages
 - ✦ directional, cheap, and easy to build
- Disadvantages
 - ✦ do not pass through solid objects
 - ✦ but also a plus
 - an infrared system in one room of a building will not interfere with a similar system in adjacent rooms or buildings

2.3.5 Light Transmission

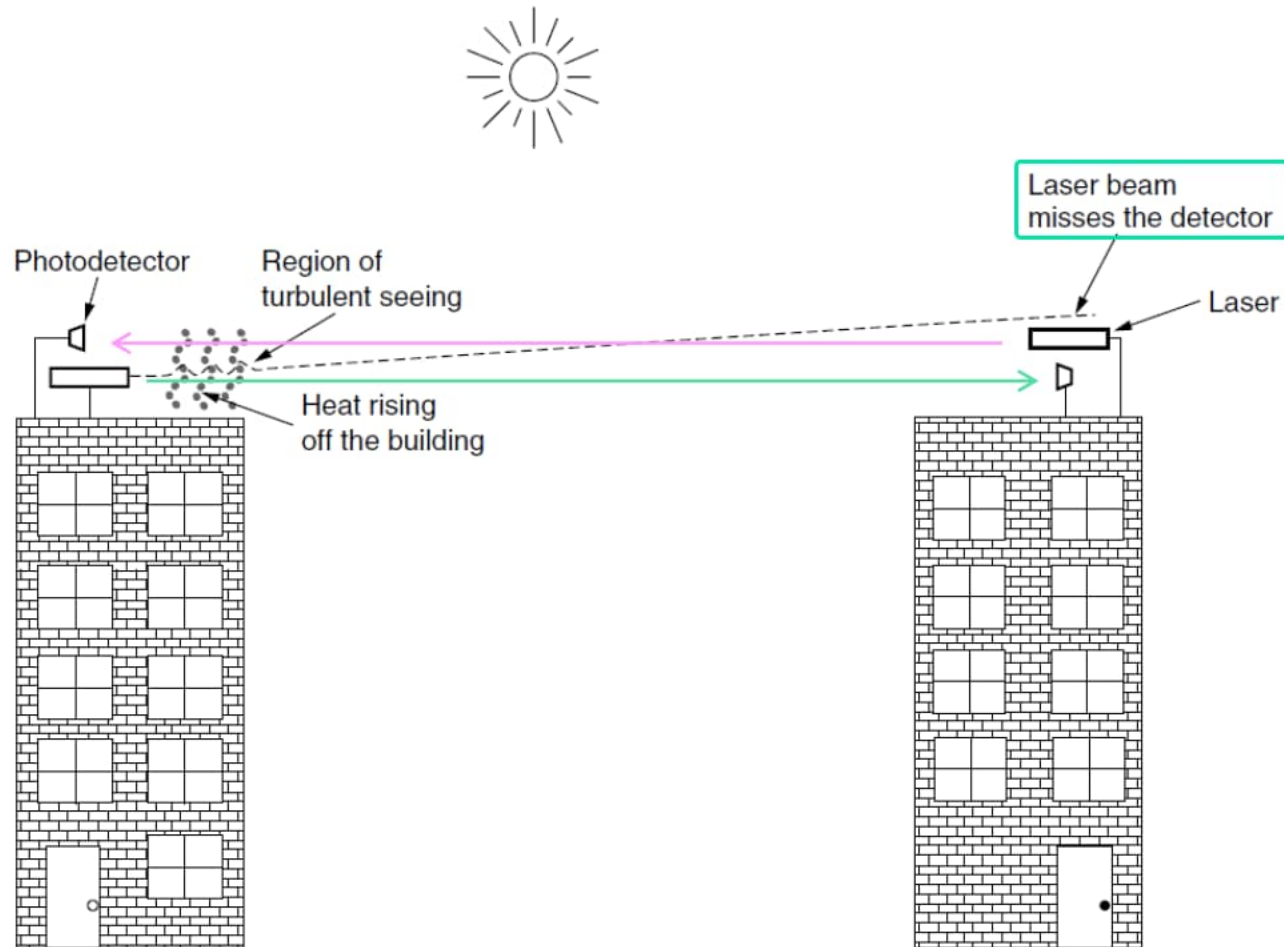


Figure 2-14. Convection currents can interfere with laser communication systems.
A **bidirectional system** with two lasers is pictured here.

convection 对流

2.4 Communication Satellites

- Artificial satellite, 人造卫星
 - ✦ can **amplify** the signals before sending them back
 - ✦ as a big **microwave repeater** in the sky.
- Regenerating signals, 信号再生
 - ✦ does not amplify noise

Van Allen belts

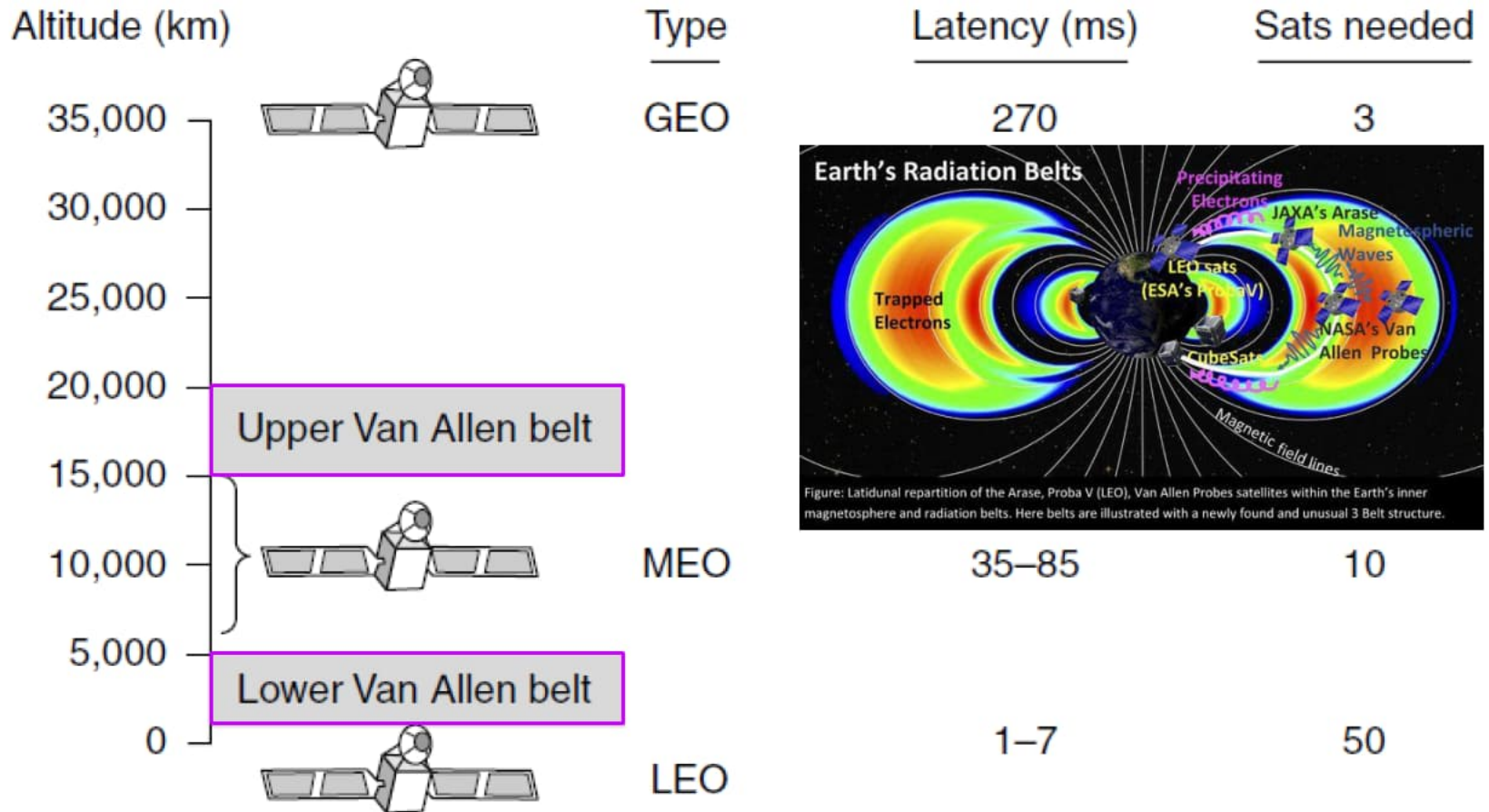


Figure 2-15. Communication satellites and some of their properties, including altitude above the earth, round-trip delay time, and number of satellites needed for global coverage.

Van Allen belts 范艾伦带, 磁层

2.4.1 Geostationary Satellites

- GEO (Geostationary Earth Orbit) satellite 地球静止轨道卫星
 - ✦ a satellite at an altitude of 35,800 km in a circular equatorial orbit

Band	Downlink	Uplink	Bandwidth	Problems
L	1.5 GHz	1.6 GHz	15 MHz	Low bandwidth; crowded
S	1.9 GHz	2.2 GHz	70 MHz	Low bandwidth; crowded
C	4.0 GHz	6.0 GHz	500 MHz	Terrestrial interference
Ku	11 GHz	14 GHz	500 MHz	Rain
Ka	20 GHz	30 GHz	3500 MHz	Rain, equipment cost

Figure 2-16. The principal satellite bands.

band 波段

- VSATs (Very Small Aperture Terminals) 甚小口径终端
 - ✦ low-cost microstations

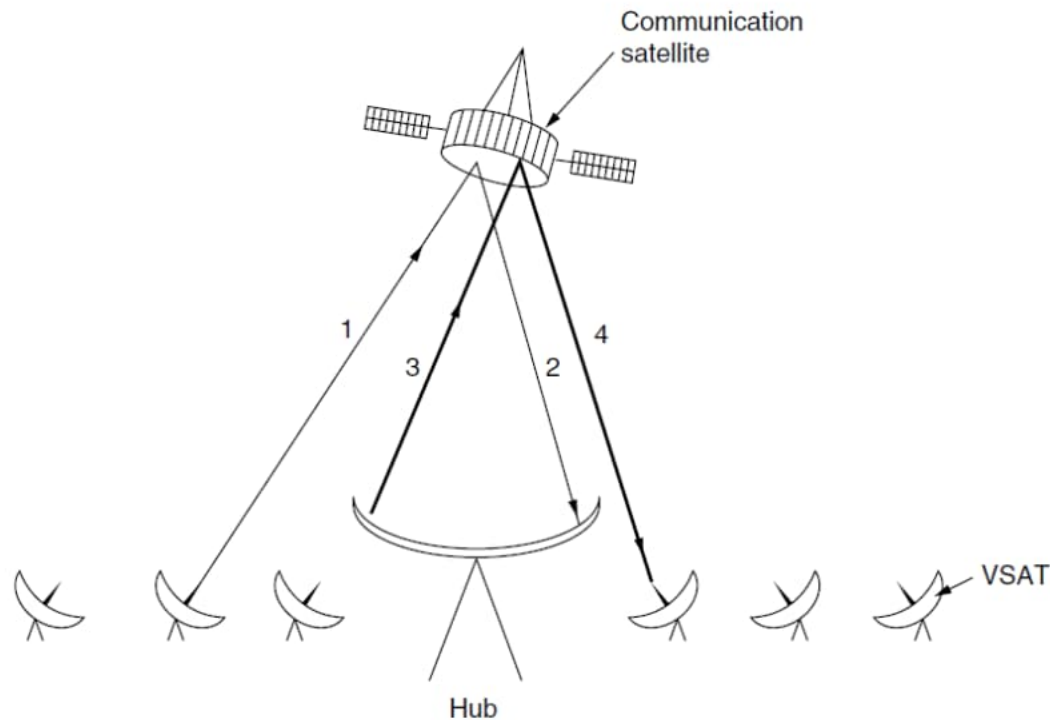


Figure 2-17. VSATs using a hub.

2.4.2 Medium-Earth Orbit Satellites



- MEO (Medium-Earth Orbit) satellites 中轨道卫星
 - ✦ between the two Van Allen belts
 - ✦ used for navigation systems rather than telecommunications
 - GPS (Global Positioning System) satellites

2.4.3 Low-Earth Orbit Satellites

- LEO (Low-Earth Orbit) satellites 低轨道卫星
 - ✦ The Iridium project, 铱星计划
 - 77 low-orbit satellites

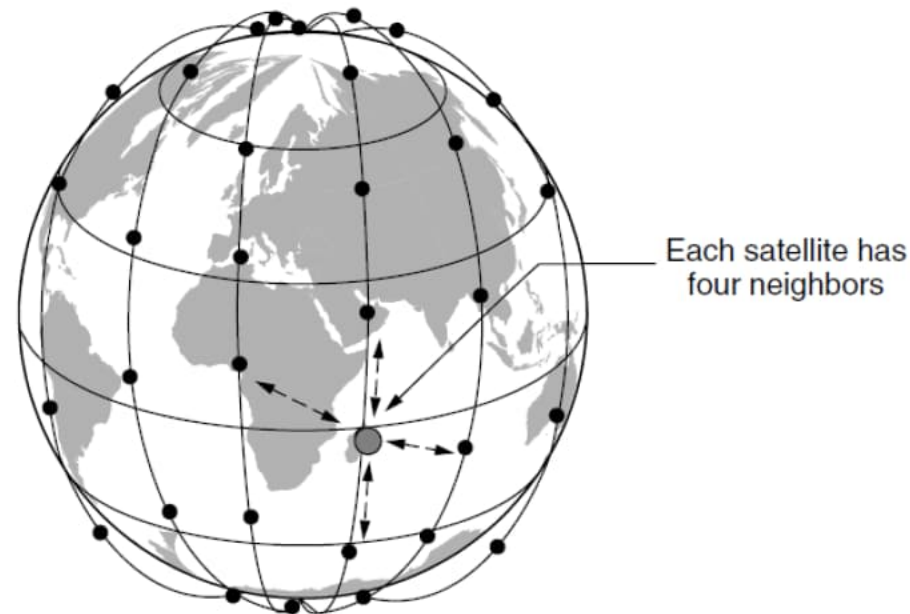
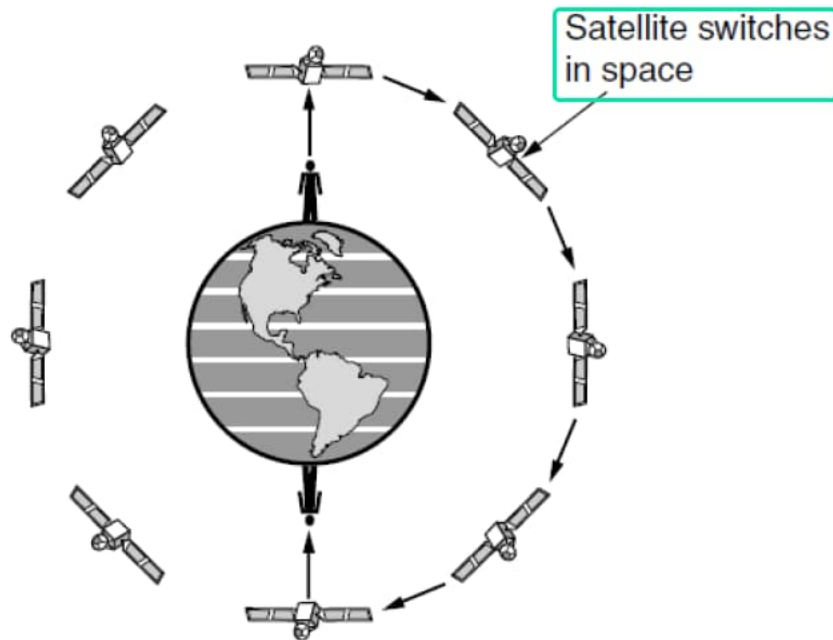
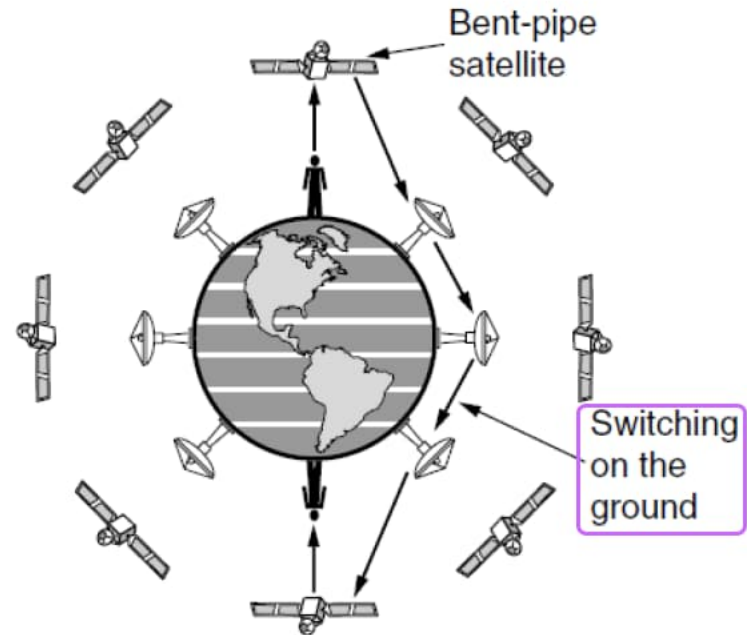


Figure 2-18. The Iridium satellites form six necklaces around the earth.

where switching



(a) Relaying in space.



(b) Relaying on the ground.

Figure 2-19.

OBP: On Board Processing, 星上处理
BP: Bent Pipe, 弯管

2.4.4 Satellites Versus Fiber

- satellites have some major niche markets
 - ★ when **rapid deployment** is critical, satellites win easily.
 - for **military** communication systems in times of war and
 - **disaster** response in times of peace
 - ★ for communication in places where **the terrestrial infrastructure is poorly** developed.
 - at the South Pole
 - Indonesia has its own satellite for domestic telephone traffic
 - among the 13,677 islands
 - ★ when **broadcasting** is essential
 - to distribute much **network TV** programming
 - transmitting a stream of **stock**, bond, or commodity prices to thousands of dealers

niche 英[nɪtʃ] 美[ni:] n. 商机; 壁龛; 合适的位置 (工作等);



2.5 Digital Modulation and Multiplexing

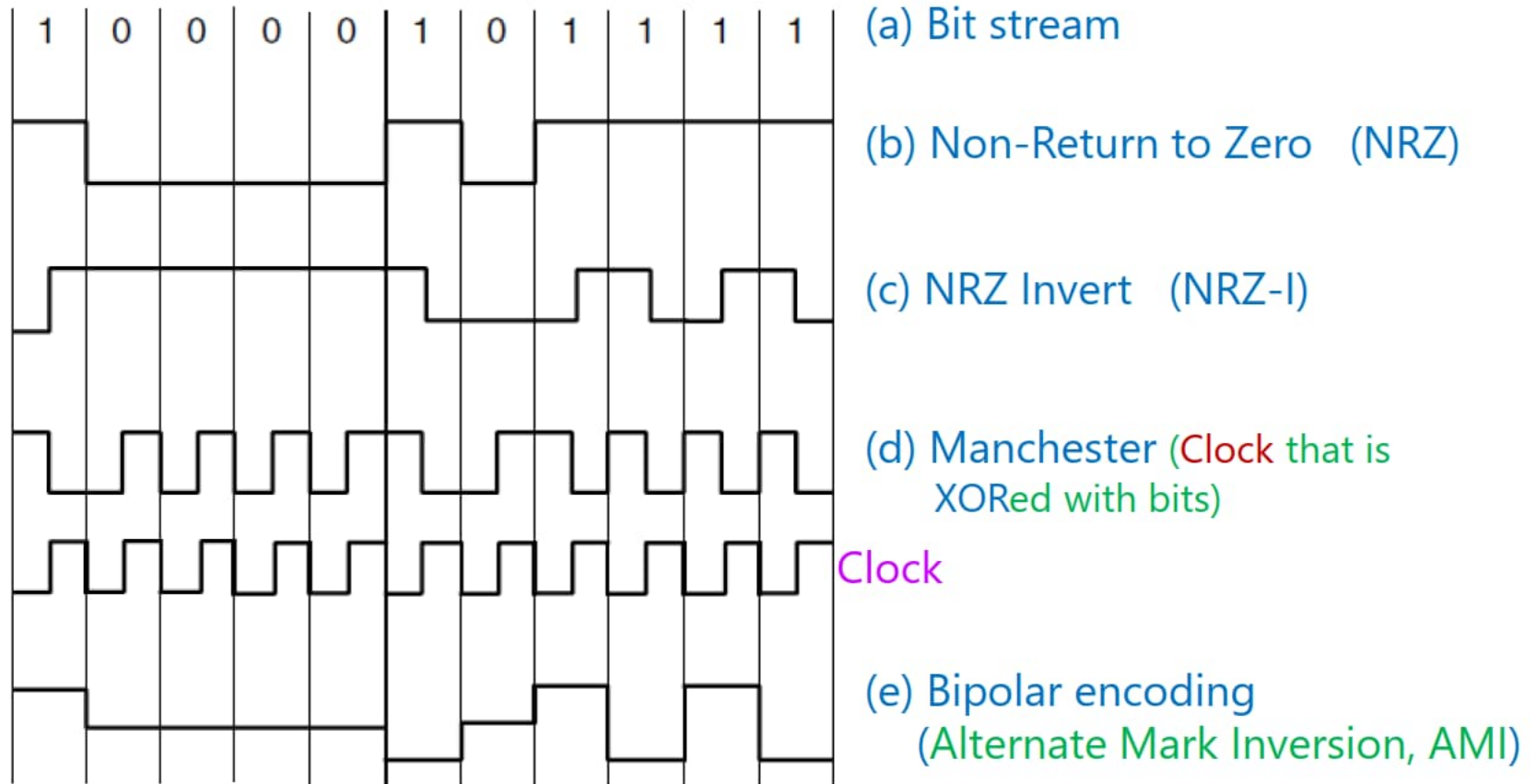
- digital modulation, 数字调制
 - ✦ The process of **converting** between **bits** and **analog** signals
- baseband transmission, 基带传输
 - ✦ the signal occupies frequencies **from zero** up to a maximum that depends on the signaling rate
- passband transmission, 频带传输
 - ✦ the signal occupies a band of frequencies **around the frequency of the carrier signal**
- multiplexing, 复用
 - ✦ to use **a single wire** to carry **several signals**
 - **time**, **frequency**, and **code** division multiplexing

2.5.1 Baseband Transmission

- Non-Return-to-Zero (NRZ), 不归零码
 - ✦ use a **positive voltage** to represent a **1** and a **negative voltage** to represent a **0**.
used in UART
- NRZ Invert (NRZI), 翻转不归零码
 - ✦ **Inverted NRZ**, used in USB2.0
- Manchester (PE: **phase encoding**), 曼切斯特编码, 相位编码
 - ✦ encoding of each data bit is **either low then high, or high then low**
 - a **self-clocking** signal with **no DC** component, used in Ethernet
- Bipolar encoding, 双极性编码
 - ✦ a type of **return-to-zero** (RZ) line code, 归零码
 - when channel is idle the line is held at a constant "**zero**" level
 - ✦ Alternate Mark Inversion (AMI), 交替传号翻转码

UART: Universal Asynchronous Receiver/Transmitter 通用异步收发报机

Line codes



Bandwidth Efficiency



- A **symbol** may be, 码元、符号
 - ✦ described as either a **pulse** in digital **baseband** transmission
 - ✦ or a **tone** in **passband** transmission using modems.
- **symbol rate**, 码元速率, 符号速率
 - ✦ also known as **baud rate** (波特率) and **modulation rate** (调制速率),
 - ✦ is the number of **symbol** changes, **waveform** changes, or **signaling** events, across the transmission medium **per time unit** using a digitally modulated signal or a line code.
- The **bit rate** is, (比特率)
 - ✦ the **symbol rate** multiplied by **the number of bits** per symbol.

4B/5B code

- Every 4 bits is mapped into a 5-bit pattern with a fixed translation table. (fast Ethernet: 4B/5B + NRZ-I)

Data (4B)	Codeword (5B)	Data (4B)	Codeword (5B)
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

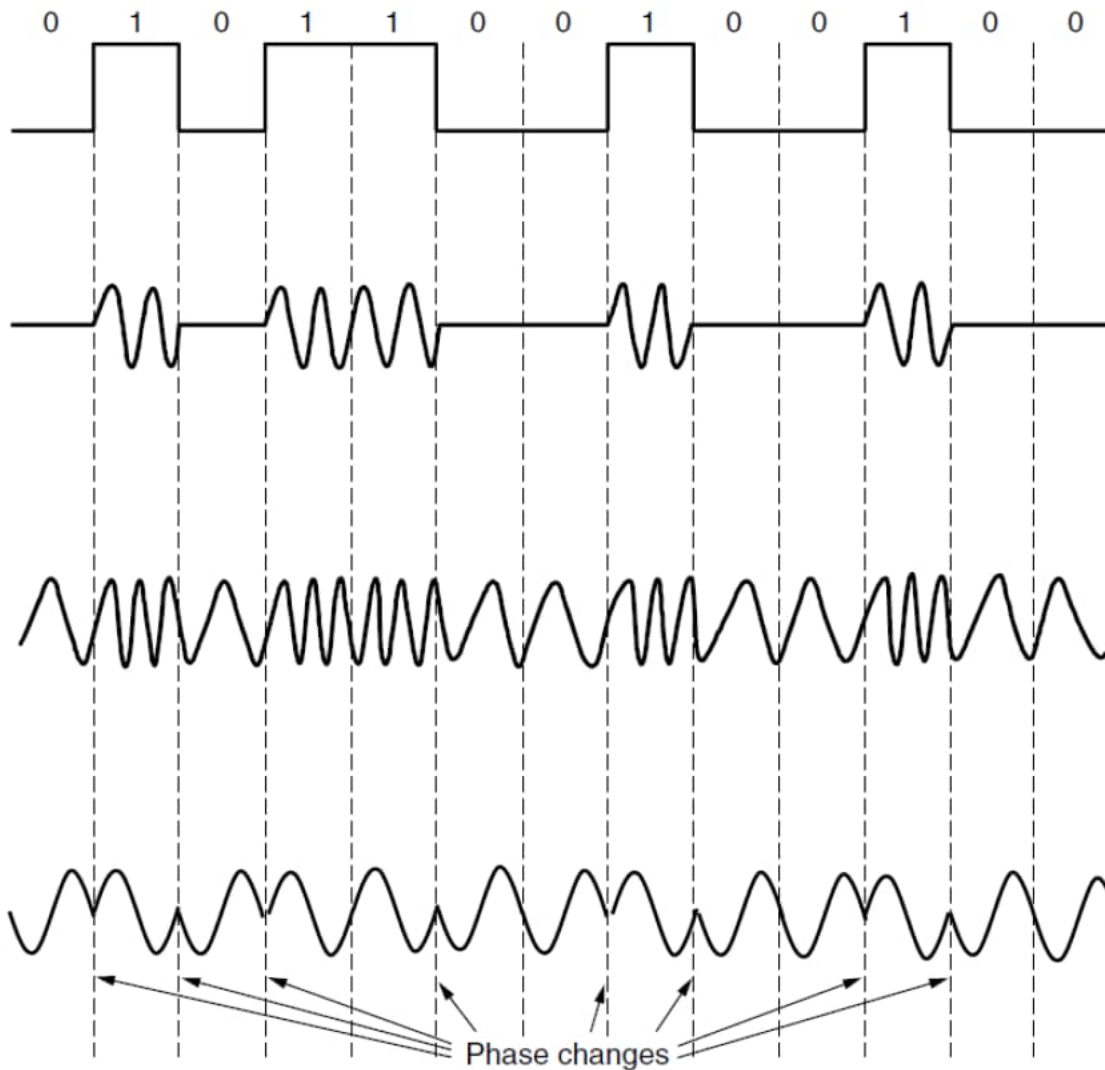
Figure 2-21. 4B/5B mapping

2.5.2 Passband Transmission

- ASK (Amplitude Shift Keying), 幅移键控
 - ✦ two different amplitudes are used to represent 0 and 1.
- FSK (Frequency Shift Keying), 频移键控
 - ✦ two or more different tones are used.
- PSK (Phase Shift Keying), 相移键控
 - ✦ BPSK (Binary Phase Shift Keying). 二进制相移键控
 - the carrier wave is systematically shifted 0 or 180 degrees at each symbol period.
 - ✦ QPSK (Quadrature Phase Shift Keying). 正交相移键控
 - to use four shifts, e.g., 45, 135, 225, or 315 degrees, to transmit 2 bits of information per symbol.

quadrature ['kwɒdrətʃə] n. 正交, 90度相位差

digital modulation



(a) A binary signal.

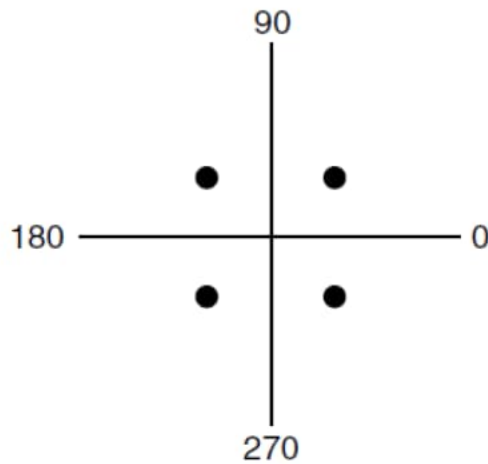
(b) Amplitude shift keying.

(c) Frequency shift keying.

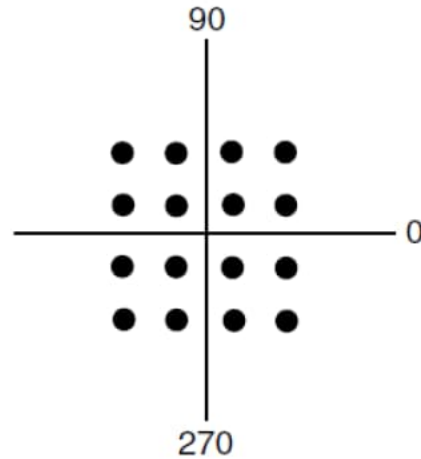
(d) Phase shift keying.

Figure 2-22

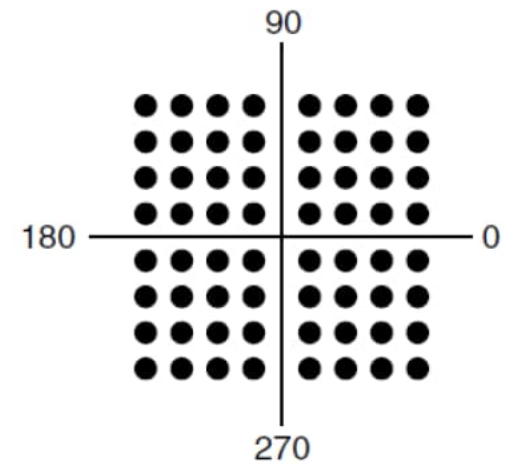
constellation diagram



(a) QPSK



(b) QAM-16



(c) QAM-64

Figure 2-23.

Constellation Diagram 星座图

Gray code

➤ Gray code, 格雷码

- ★ to map bits to symbols so that adjacent symbols differ in only 1 bit position

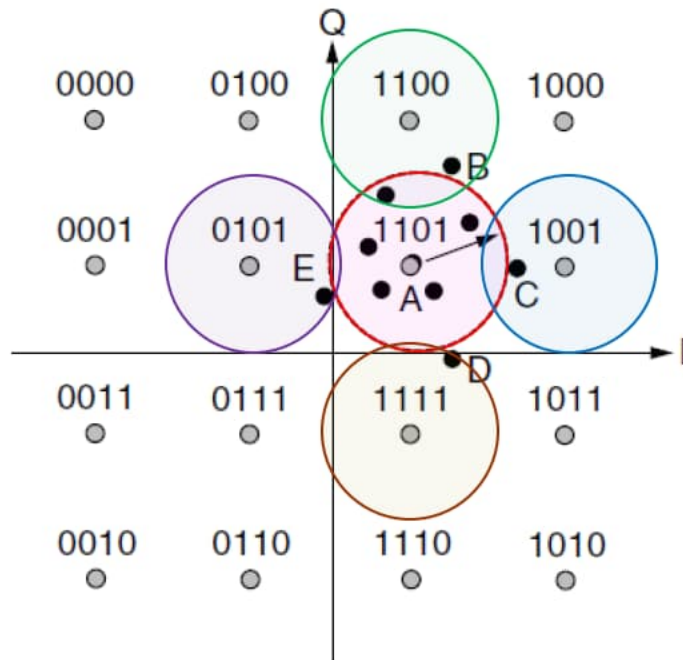


Figure 2-24. Gray-coded QAM-16.

When 1101 is sent:

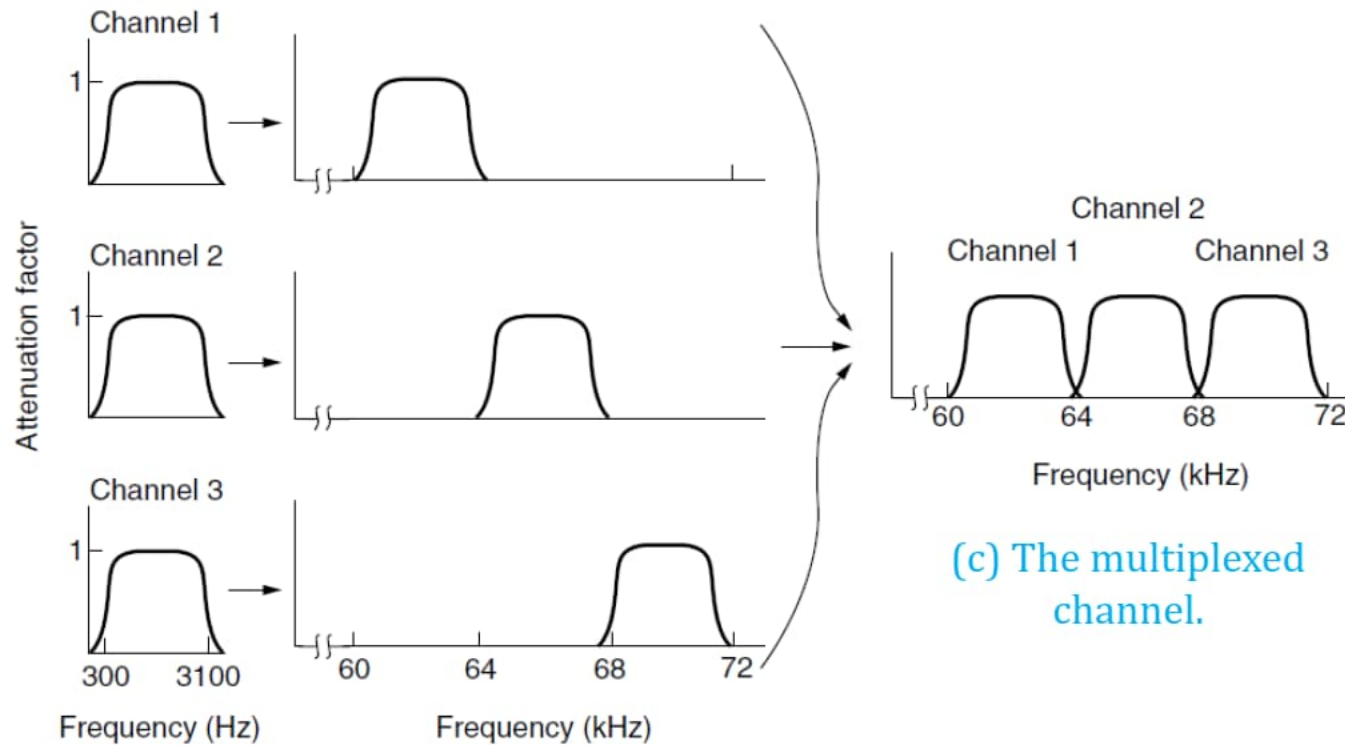
Point	Decodes as	Bit errors
A	1101	0
B	110 <u>0</u>	1
C	<u>1</u> 001	1
D	11 <u>1</u> 1	1
E	<u>0</u> 101	1

Hamming distance = 1

2.5.3 Frequency Division Multiplexing



➤ FDM (Frequency Division Multiplexing), 频分复用



(a) The original bandwidths.

(b) The bandwidths raised in frequency.

(c) The multiplexed channel.

Figure 2-25. Frequency division multiplexing.

➤ OFDM

★ Orthogonal frequency division multiplexing, 正交频分复用

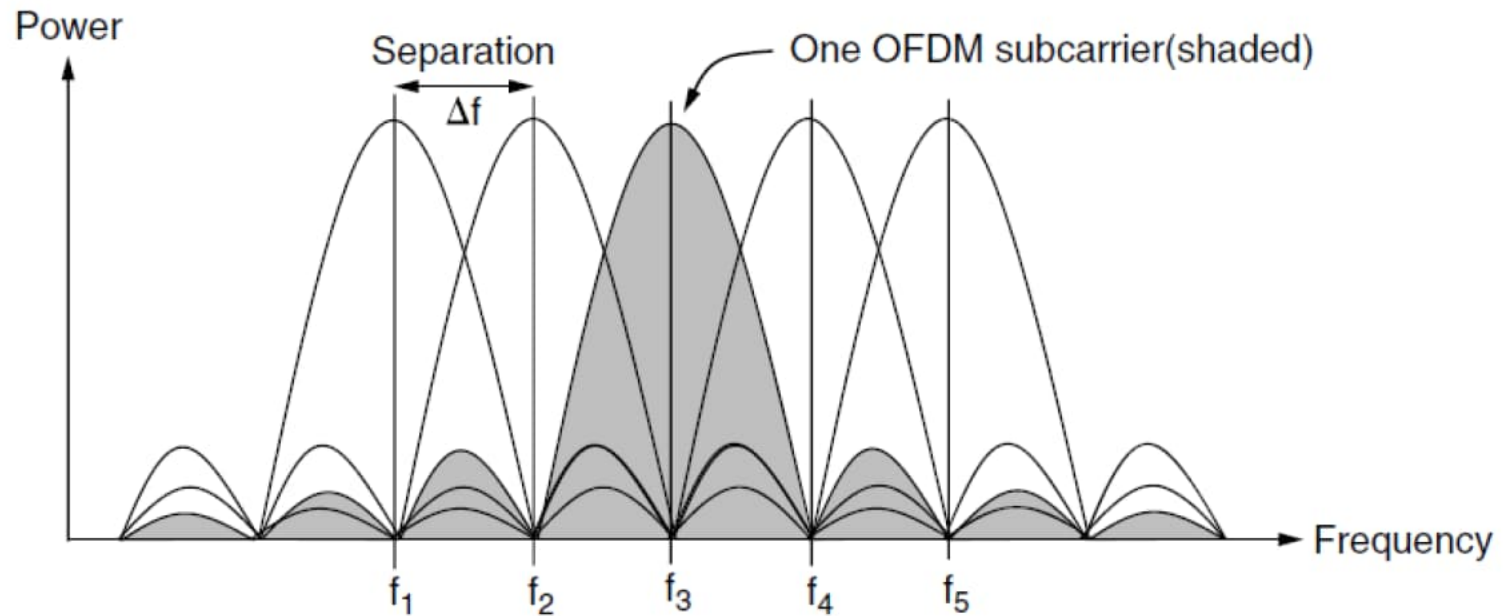


Figure 2-26. Orthogonal frequency division multiplexing (OFDM).

Orthogonal [ɔ:'θɒɡənəl] adj. 正交的; (相互) 垂直的;

2.5.4 Time Division Multiplexing

- TDM (Time Division Multiplexing), 时分复用
 - ✦ time slot, 时隙

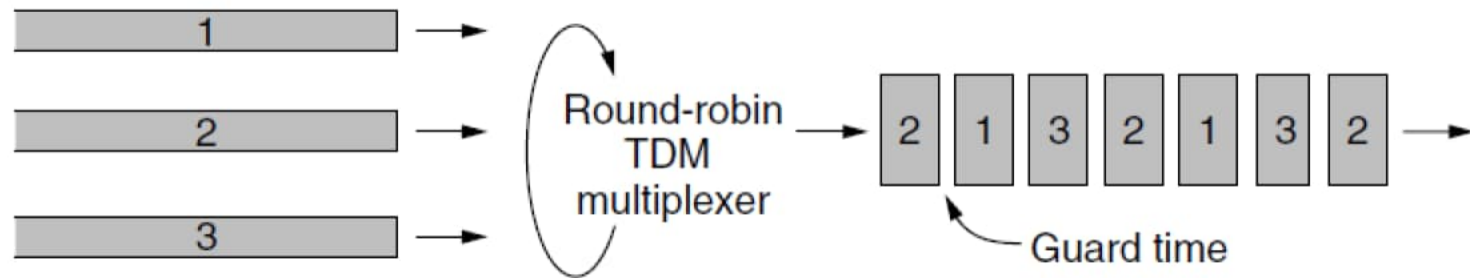


Figure 2-27. Time Division Multiplexing (TDM).

Round-robin 轮询

2.5.5 Code Division Multiplexing

- CDM, 码分复用
 - ✦ Code Division Multiplexing
- CDMA, 码分多址
 - ✦ Code Division Multiple Access
- chip, 码片
 - ✦ In CDMA, each bit time is subdivided into m short intervals
- chip sequence, 码片序列
 - ✦ Each station is assigned a unique m -bit code
- orthogonal, 正交
 - ✦ the normalized inner product of any two distinct chip sequences, $\mathbf{S}$ and $\mathbf{T}$ (written as $\mathbf{S} \cdot \mathbf{T}$), is 0.

$$\mathbf{S} \cdot \mathbf{T} \equiv \frac{1}{m} \sum_{i=1}^m S_i T_i = 0$$

signal recovery



$$A = (-1 \ -1 \ -1 \ +1 \ +1 \ -1 \ +1 \ +1)$$

$$B = (-1 \ -1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1)$$

$$C = (-1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1 \ -1)$$

$$D = (-1 \ +1 \ -1 \ -1 \ -1 \ -1 \ +1 \ -1)$$

(a) Chip sequences for four stations.

$$S1=C = (-1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1 \ -1)$$

$$S2=B+C = (-2 \ 0 \ 0 \ 0 \ +2 \ +2 \ 0 \ -2)$$

$$S3=A+!B = (0 \ 0 \ -2 \ +2 \ 0 \ -2 \ 0 \ +2)$$

$$S4=A+!B+C = (-1 \ +1 \ -3 \ +3 \ +1 \ -1 \ -1 \ +1)$$

$$S5=A+B+C+D = (-4 \ 0 \ -2 \ 0 \ +2 \ 0 \ +2 \ -2)$$

$$S6=A+B+!C+D = (-2 \ -2 \ 0 \ -2 \ 0 \ -2 \ +4 \ 0)$$

(c) Six examples of transmissions.

$$S1 \bullet C = [1+1+1+1+1+1+1+1]/8 = 1$$

$$S2 \bullet C = [2+0+0+0+2+2+0+2]/8 = 1$$

$$S3 \bullet C = [0+0+2+2+0-2+0-2]/8 = 0$$

$$S4 \bullet C = [1+1+3+3+1-1+1-1]/8 = 1$$

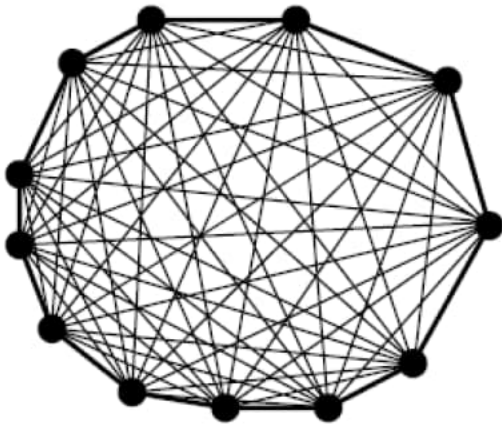
$$S5 \bullet C = [4+0+2+0+2+0-2+2]/8 = 1$$

$$S6 \bullet C = [2-2+0-2+0-2-4+0]/8 = -1$$

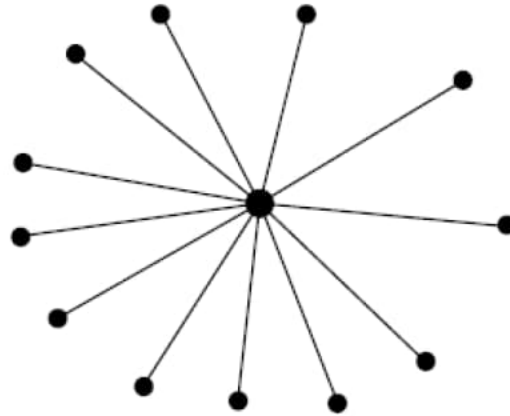
书本有错

(d) Recovery of station C's signal.

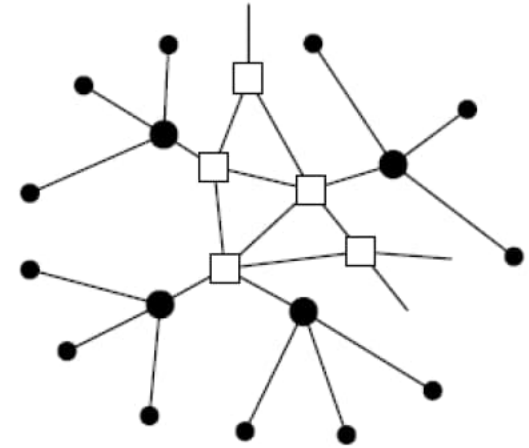
2.6 The Public Switched Telephone Network



(a) Fully interconnected network.



(b) Centralized switch.



(c) Two-level hierarchy.

Figure 2-29.

Terminologies



- PSTN, 公共交换电话网
 - ✦ Public Switched Telephone Network
- call, 呼叫
- switch office, 交换局
 - ✦ where calls are moved from one trunk to another
- trunk, 中继线
 - ✦ digital fiber optic links connecting the switching offices
- end office (local central office), 端局
- local loop, 本地回路
 - ✦ analog twisted pairs going to houses and businesses
- toll office, 长途局
 - ✦ tandem office (if they are within the same local area), 汇接局

long-distance call

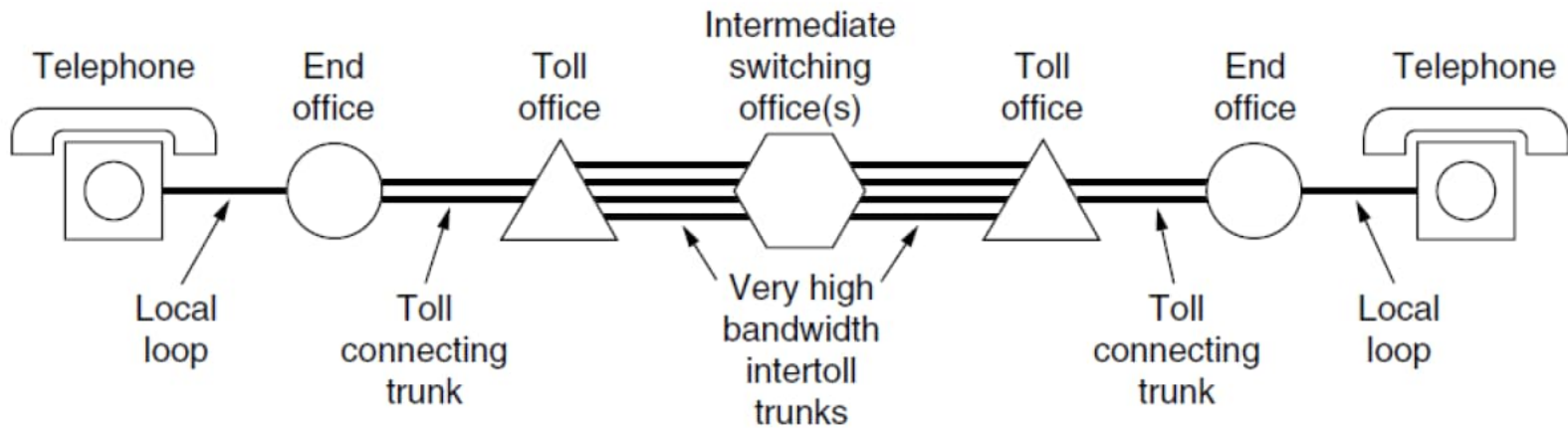


Figure 2-30. A typical circuit route for a long-distance call.

Telephone Modems

- modem, 调制解调器
 - ✦ is short for “**mod**ulator **dem**odulator.”
- codec, 编码解码器, 声码器
 - ✦ short for “**cod**er-**dec**oder”

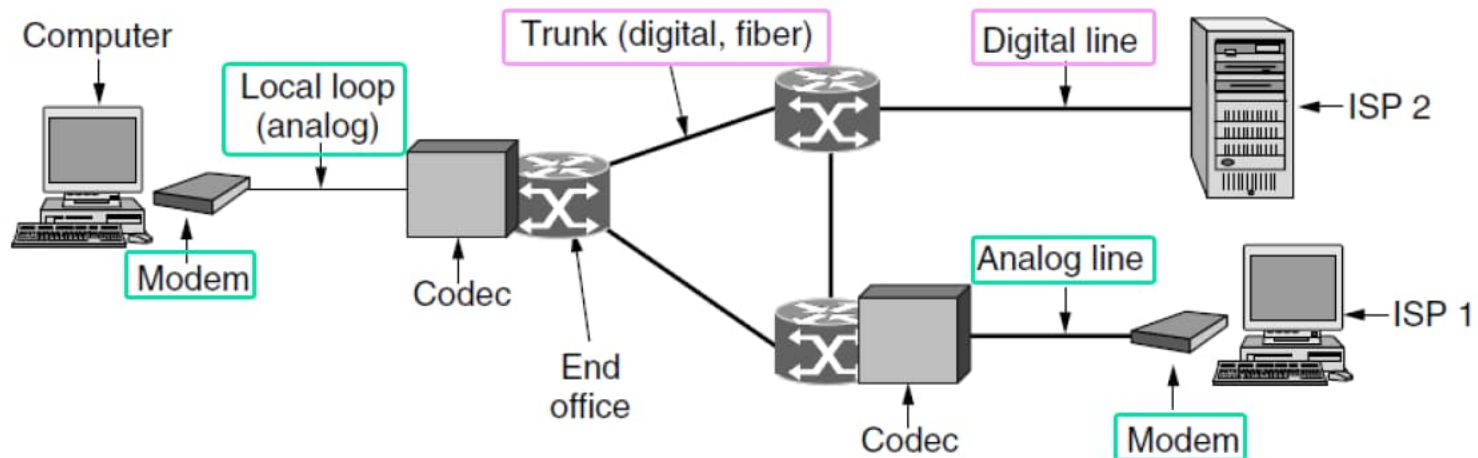


Figure 2-32. The use of both **analog** and **digital** transmission for a computer-to-computer call. Conversion is done by the modems and codecs.

ADSL



➤ ADSL (Asymmetric Digital Subscriber Line)

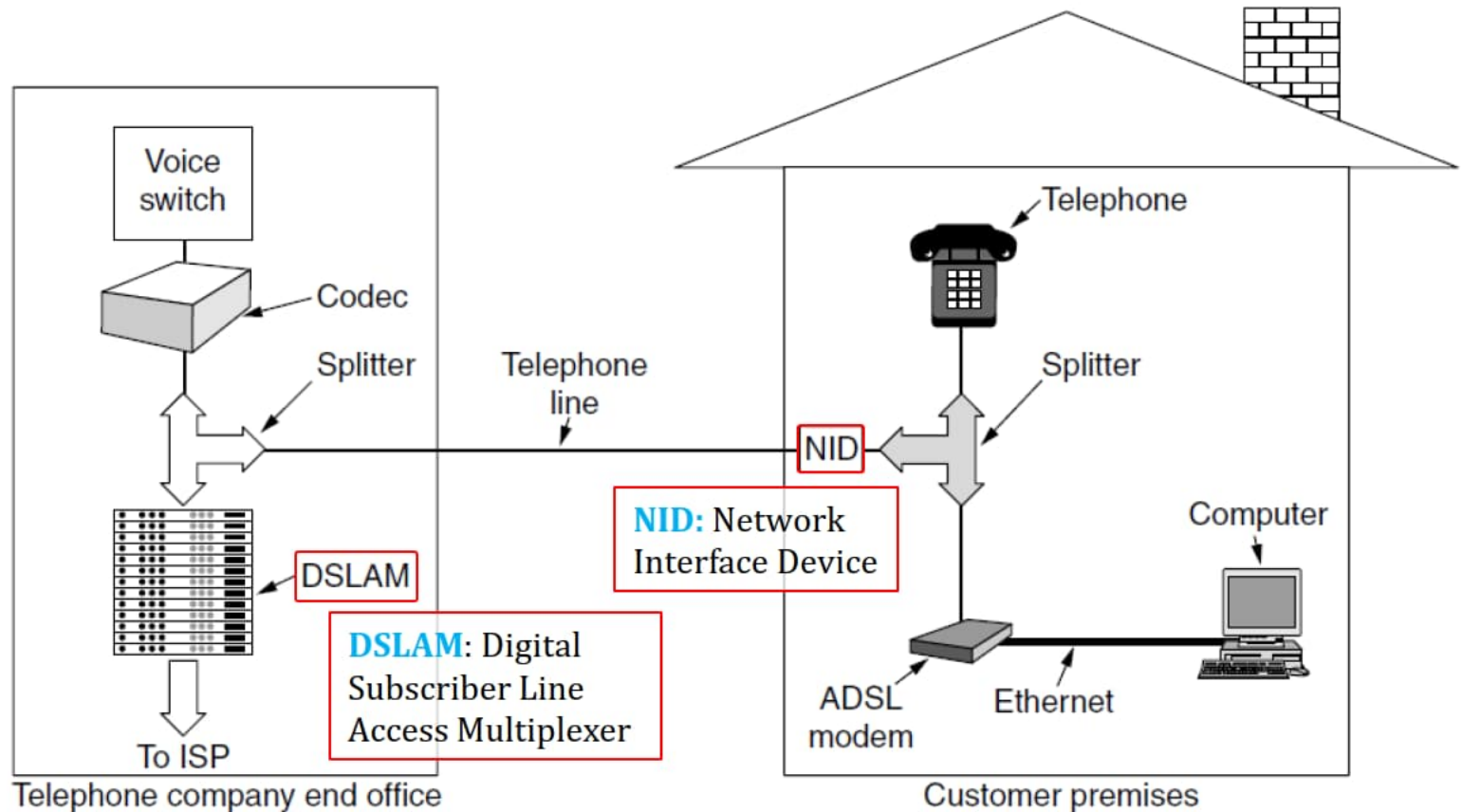


Figure 2-35. A typical ADSL equipment configuration.

FTTH: Fiber To The Home



- PON (Passive Optical Network), 无源光网络
 - ✦ GPON (Gigabit-capable PON), from ITU
 - ✦ EPON (Ethernet PON), from IEEE

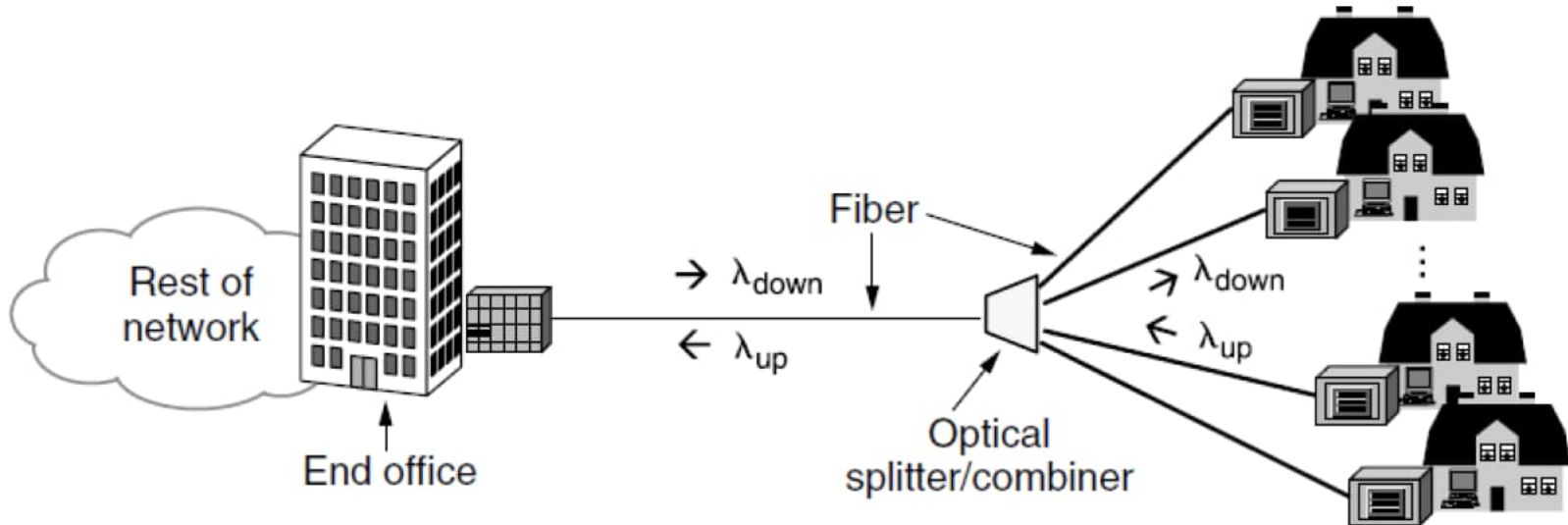


Figure 2-36. Passive optical network for Fiber To The Home.

Digitizing Voice Signals



- group, 基群
 - ✦ 12 calls in the 60 kHz–to–108 kHz band
- supergroup, 超群
 - ✦ five groups (a total of 60 calls)
- codec (short for “coder-decoder”), 编码解码器, 声码器
 - ✦ a device digitizing the analog signals in the end office
- PCM (Pulse Code Modulation), 脉码调制
 - ✦ time intervals: $1 / (8\text{KHz}) = 125 \mu\text{sec}$
 - ✦ μ – law, used in North America and Japan, μ 律
 - ✦ A – law, used in Europe and the rest of the world, A律

Time Division Multiplexing

➤ T1 carrier, T1载波

- ★ TDM scheme used in North America and Japan
- ★ $(24 \times 8 + 1) \text{ bits} / 125 \mu\text{sec} = 1.544 \text{ Mbps}$

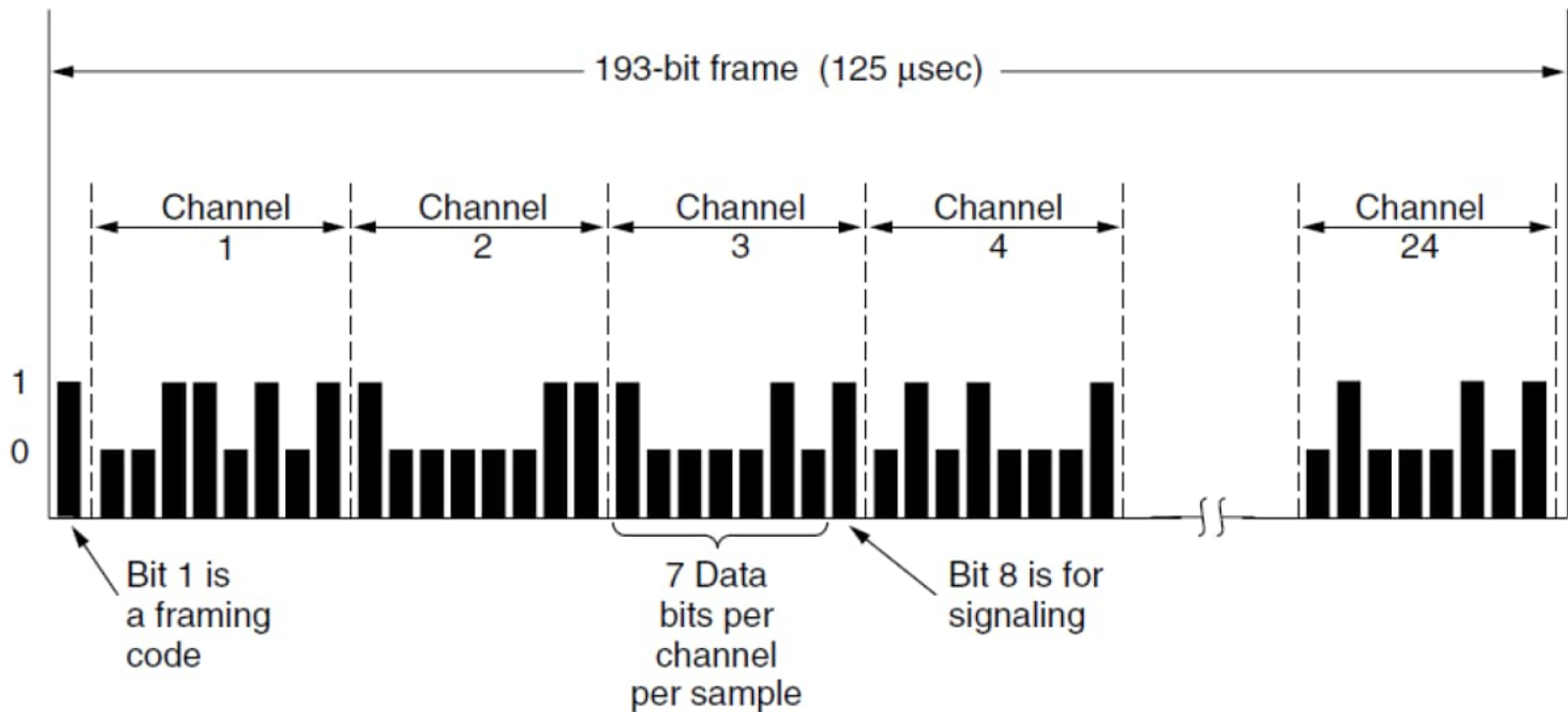


Figure 2-37. The T1 carrier (1.544 Mbps).

Multiplexing T1

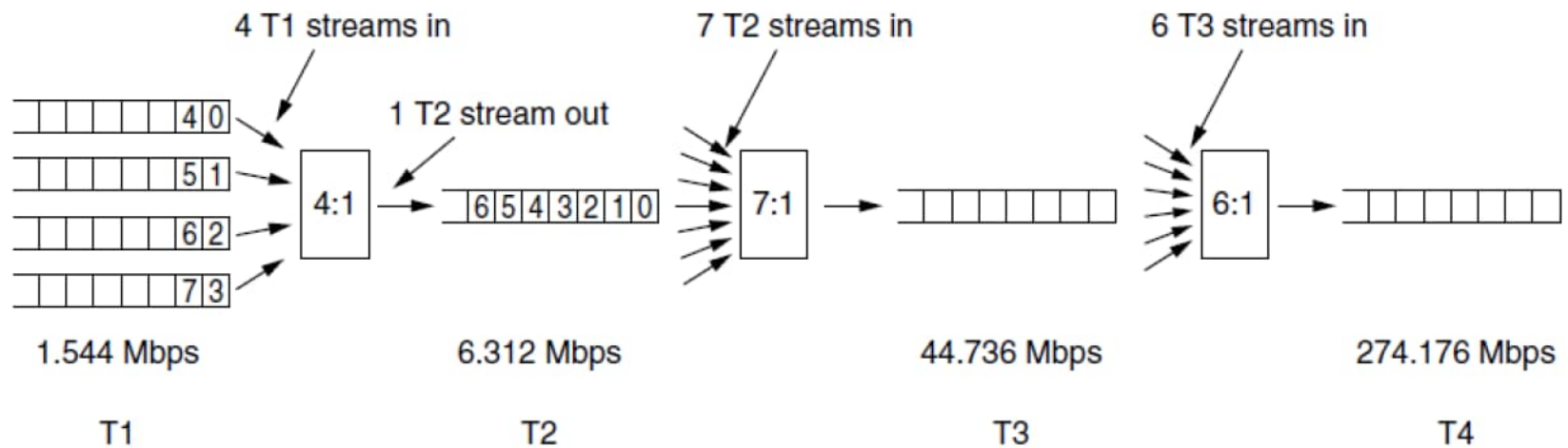


Figure 2-38. Multiplexing T1 streams into higher carriers.

E1 carrier

➤ E1 carrier, E1载波

- ✦ Outside North America and Japan
- ✦ 32 8-bit data samples packed into the basic 125-μsec frame
 - Thirty of the channels are used for information and up to two are used for signaling
- ✦ $32 * 8 \text{ bits} / 125 \mu\text{sec} = 2.048 \text{ Mbps}$

➤ Multiplexing E1

- ✦ $E2 = E1 * 4 = 8.848 \text{ Mbps}$
- ✦ $E3 = E2 * 4 = 34.304 \text{ Mbps}$
- ✦ $E4 = E3 * 4 = 139.264 \text{ Mbps}$

SONET/SDH



- SONET/SDH, 同步光网络/同步数字系列
 - ✦ SONET: Synchronous Optical Network, from Bellcore, (Bell Lab Communication Research)
 - ✦ SDH: Synchronous Digital Hierarchy, from ITU
 - ✦ (omitted)

Wavelength Division Multiplexing



- WDM, 波分复用
 - ✦ Wavelength Division Multiplexing
- DWDM (Dense WDM), 密集波分复用

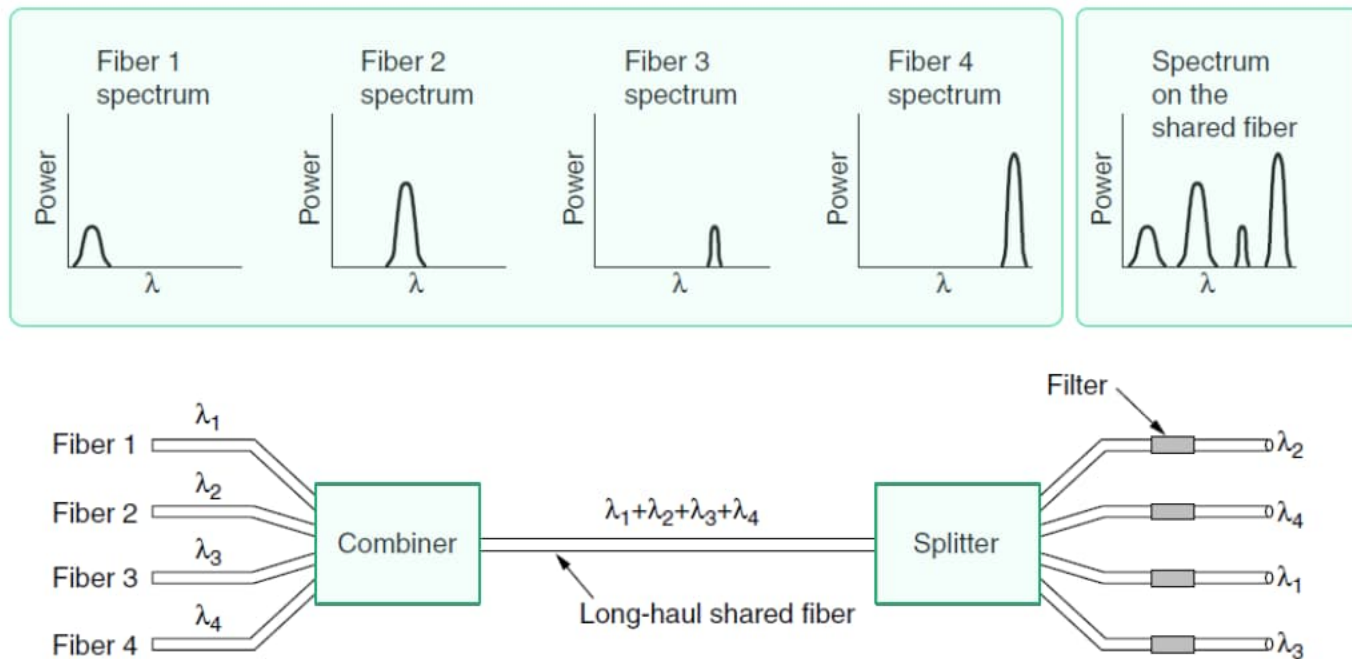


Figure 2-41. Wavelength division multiplexing.

2.6.5 Switching



➤ Circuit Switching, 电路交换

- ★ the switching equipment within the telephone system seeks out a **physical path** all the way from your telephone to the receiver's telephone

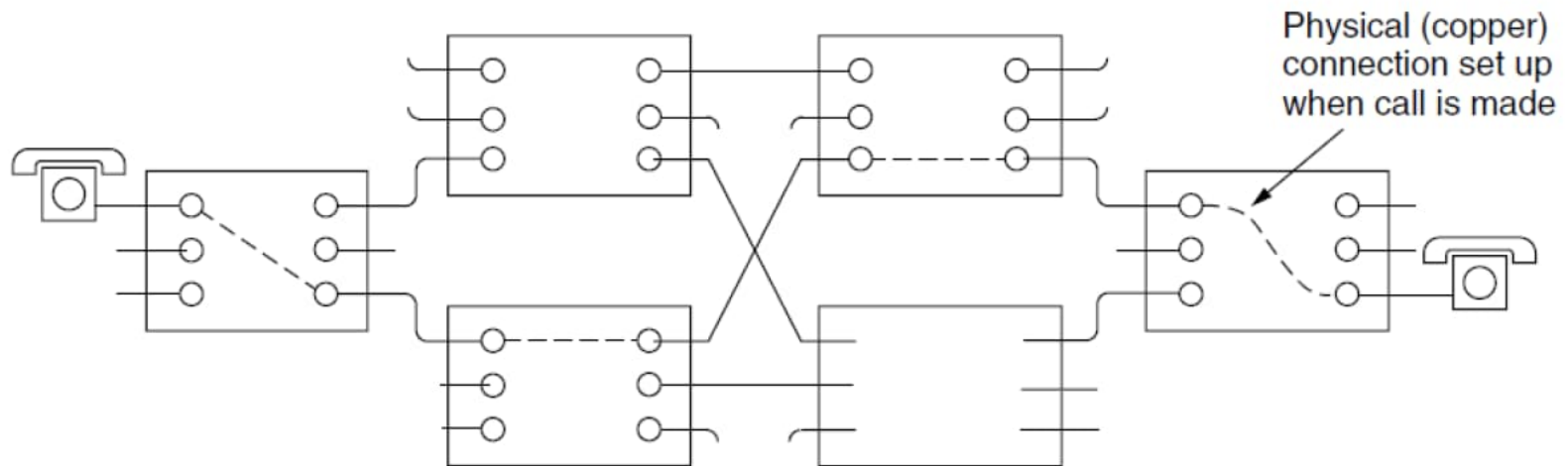


Figure 2-42. (a) Circuit switching.

➤ Packet Switching, 分组交换

- ✦ There is **no need** to **set up** a dedicated path in advance
- ✦ packets are sent as soon as they are available

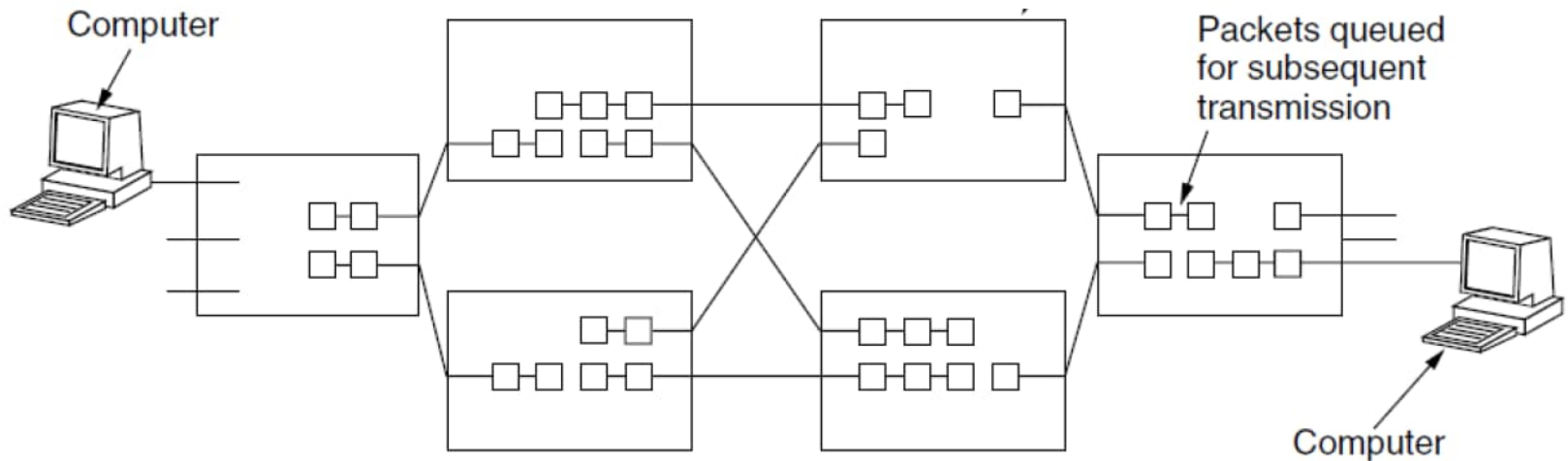
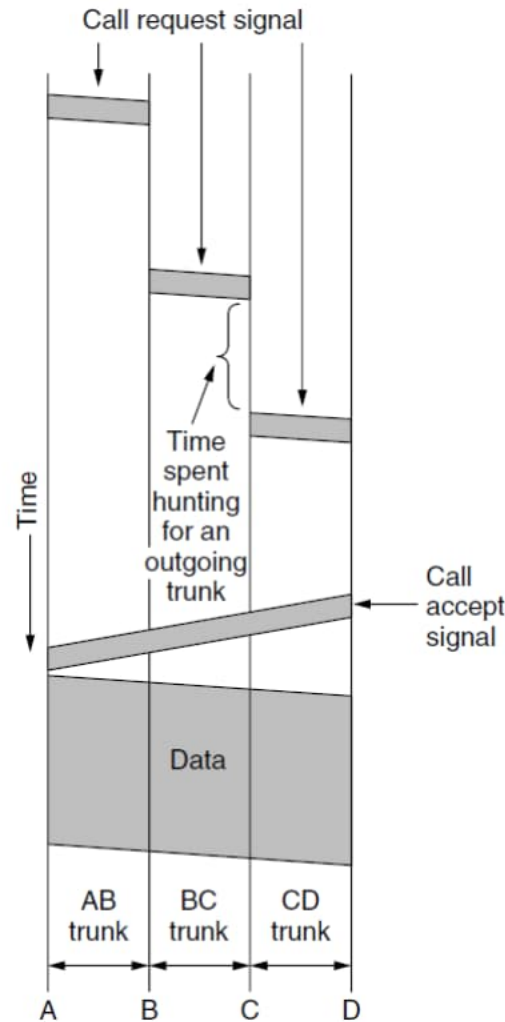


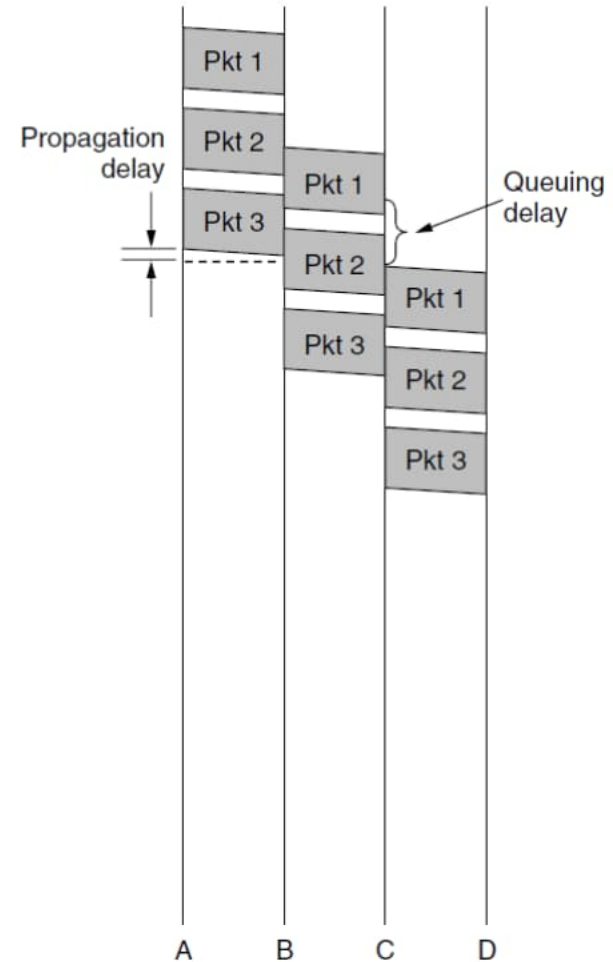
Figure 2-42. (b) Packet switching.

Timing of events

Figure 2-43. Timing of events in
(a) circuit switching,
(b) packet switching.



(a) circuit switching



(b) packet switching

Item	Circuit switched	Packet switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
Time of possible congestion	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Charging	Per minute	Per packet

Figure 2-44. A comparison of circuit-switched and packet-switched networks.

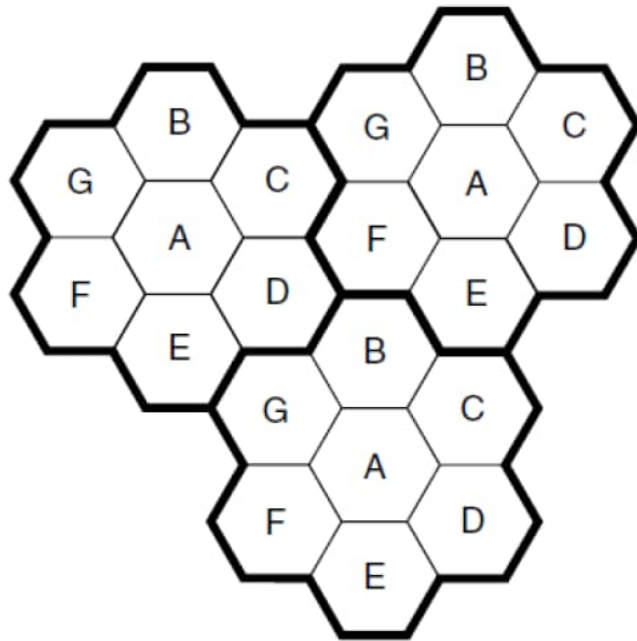
2.7 The Mobile Telephone System



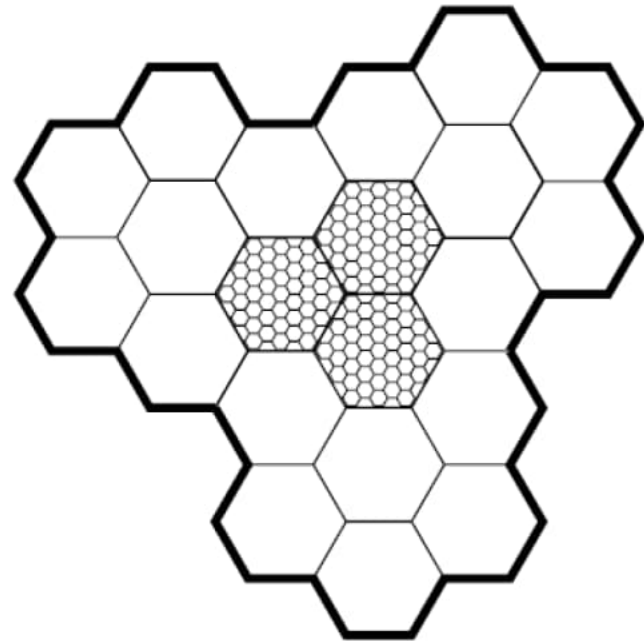
- Mobile phones (sometimes called cell phones)
 - ✦ 1G. Analog voice.
 - ✦ 2G. Digital voice.
 - ✦ 3G. Digital voice and data (Internet, email, etc.).

2.7.1 First-Generation (1G) Mobile Phones: Analog Voice

- AMPS (Advanced Mobile Phone System),
 - ✦ invented by **Bell Labs** and first installed in the United States in **1982**.



(a) Frequencies are not reused in adjacent cells.



(b) To add more users, smaller cells can be used.

Figure 2-45.

2.7.2 Second-Generation (2G) Mobile Phones: Digital Voice

- GSM (Global System for Mobile communications)

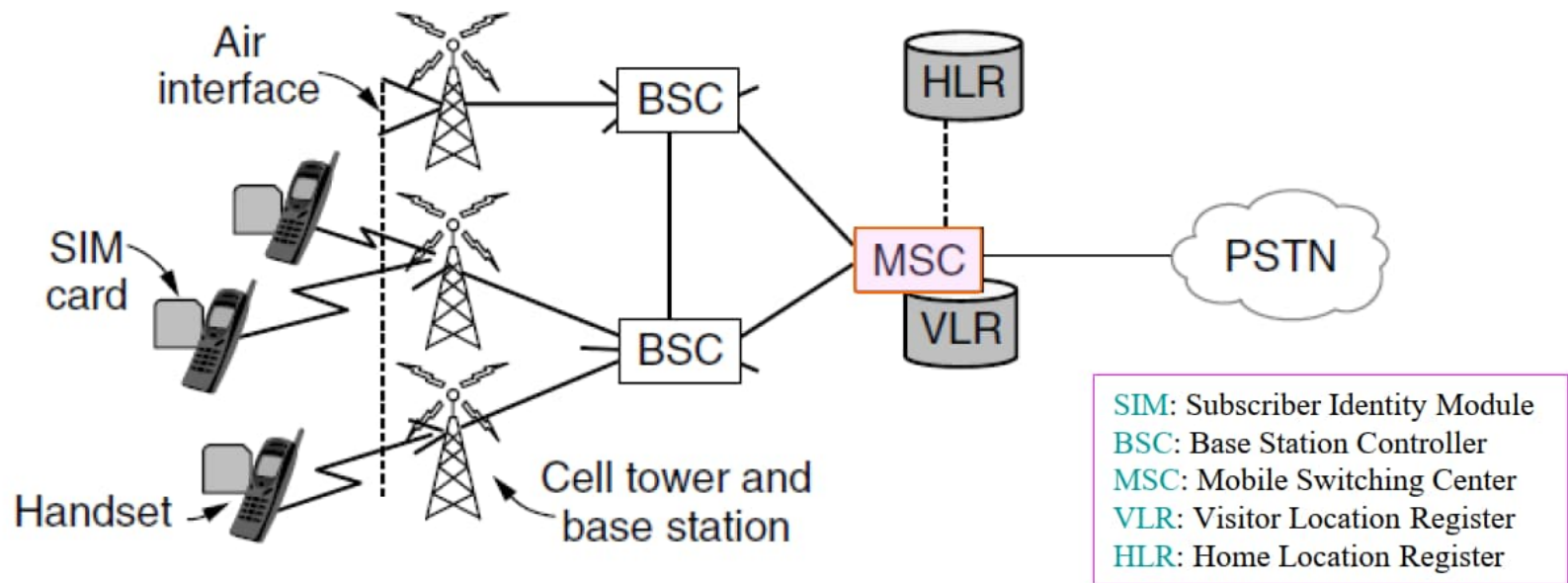


Figure 2-46. GSM mobile network architecture.

Channels

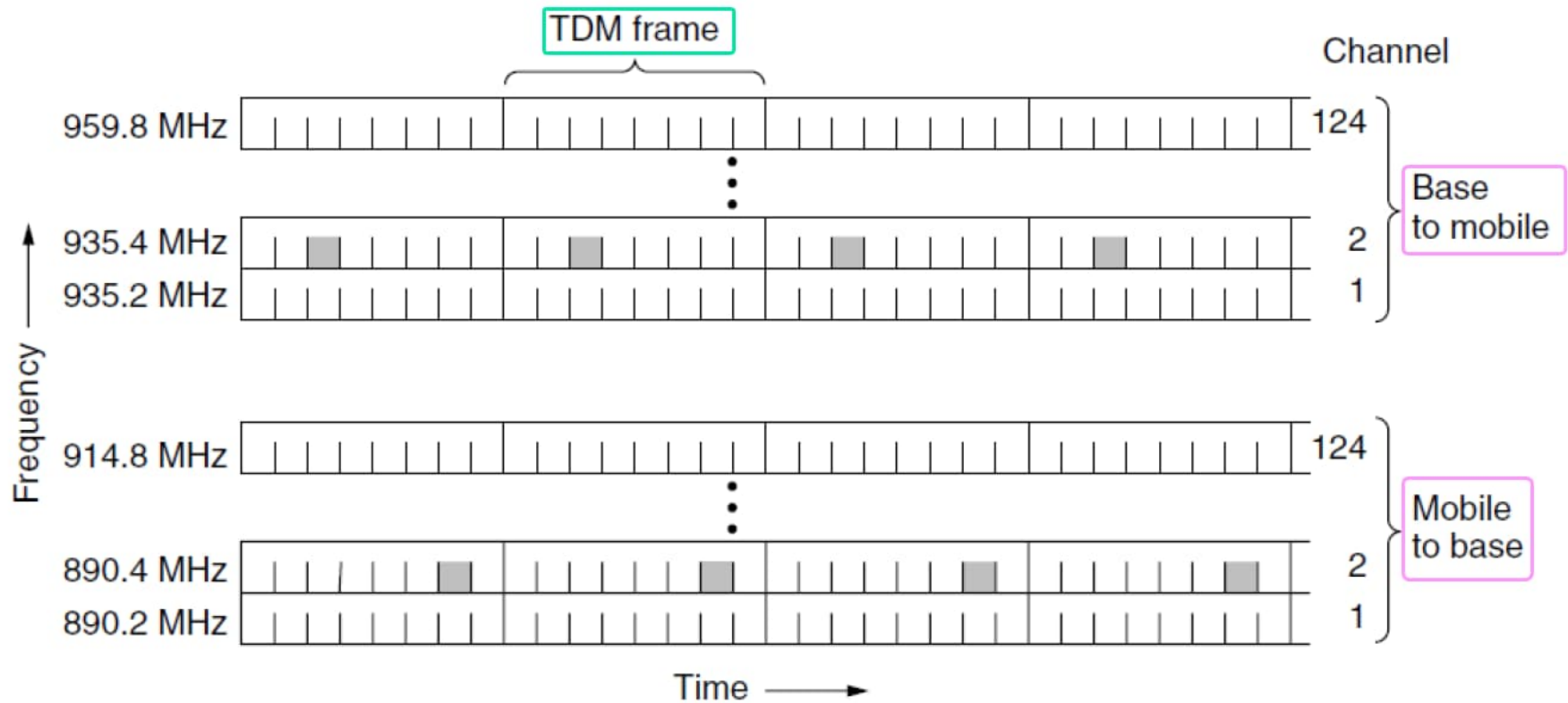


Figure 2-47. GSM uses 124 frequency channels, each of which uses an eight-slot TDM system.



➤ IMT-2000

- ✦ International Mobile Telecommunications
- ✦ WCDMA (Wideband CDMA),
 - UMTS (Universal Mobile Telecommunications System).
 - was proposed by Ericsson and was pushed by the European Union
- ✦ CDMA2000,
 - proposed by Qualcomm.
- ✦ TD-SCDMA
 - proposed by DTT (Motorola) and was pushed by China-Mobile.

2.8 Cable Television

Drop cable: 分支电缆

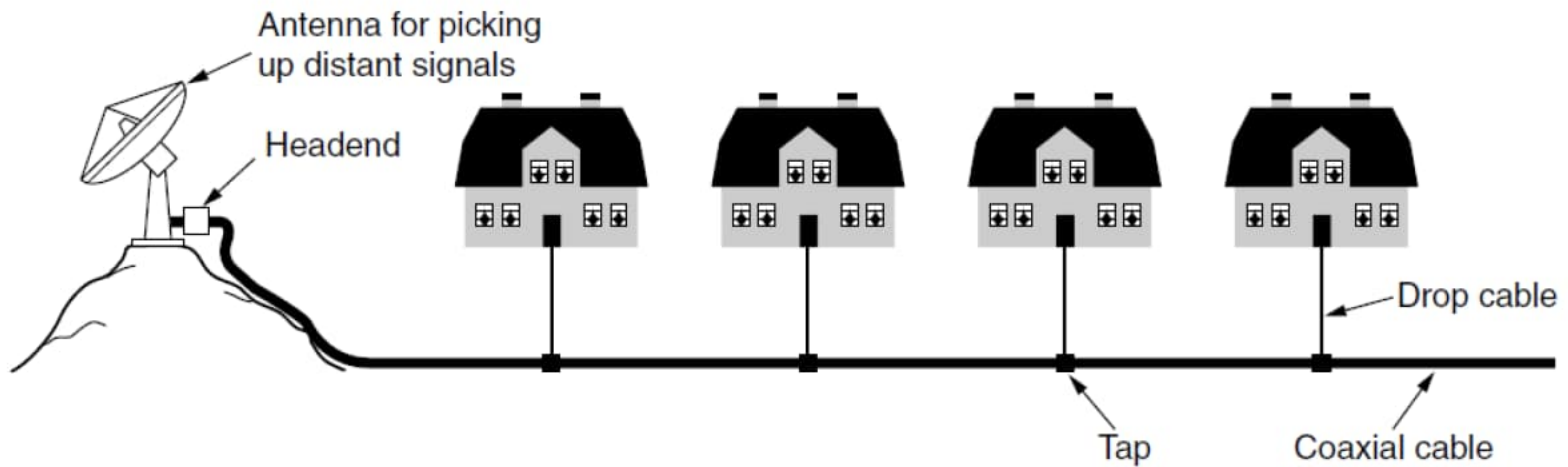


Figure 2-50. An early cable television system.

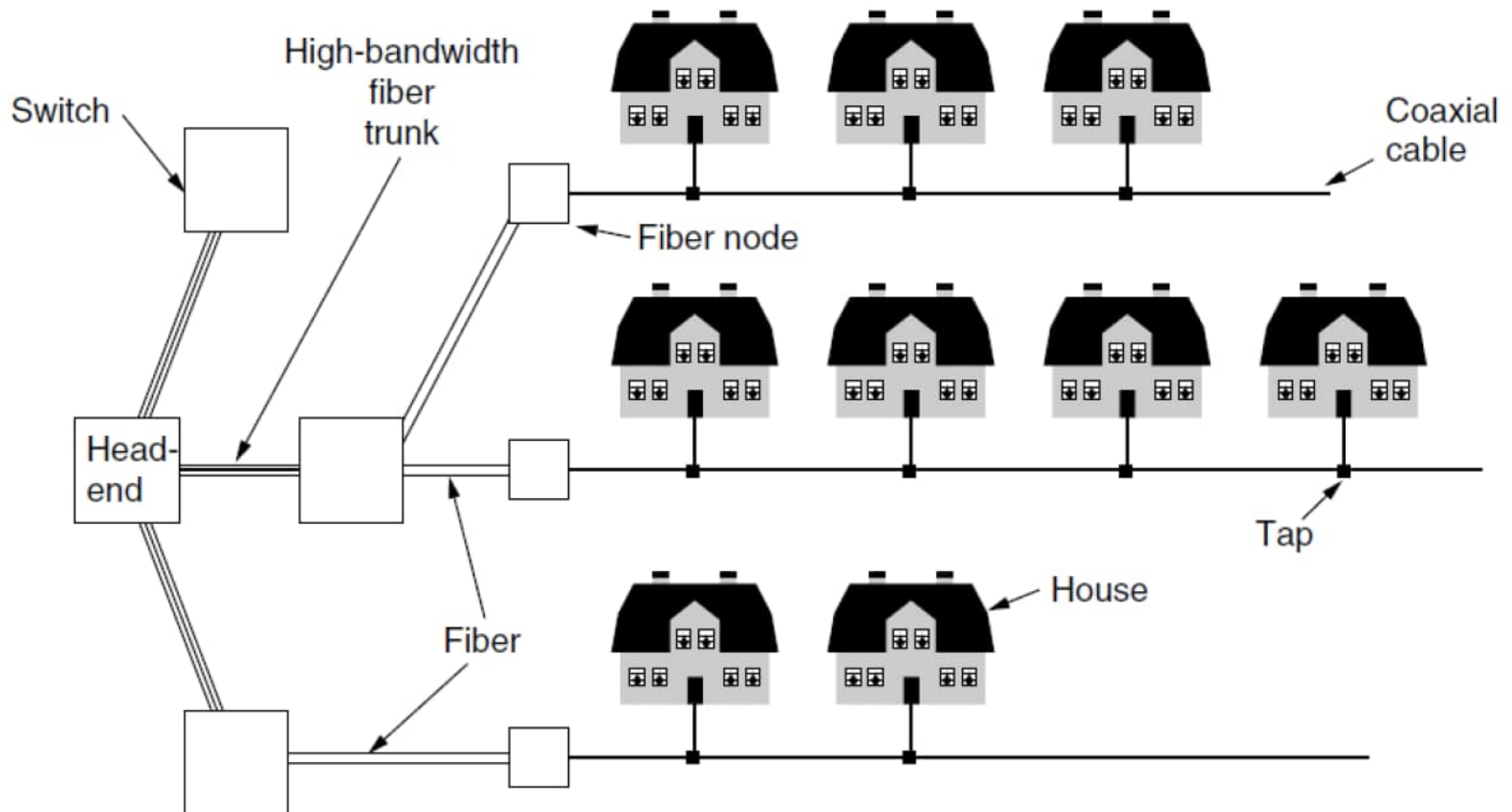


Figure 2-51. (a) Cable television.

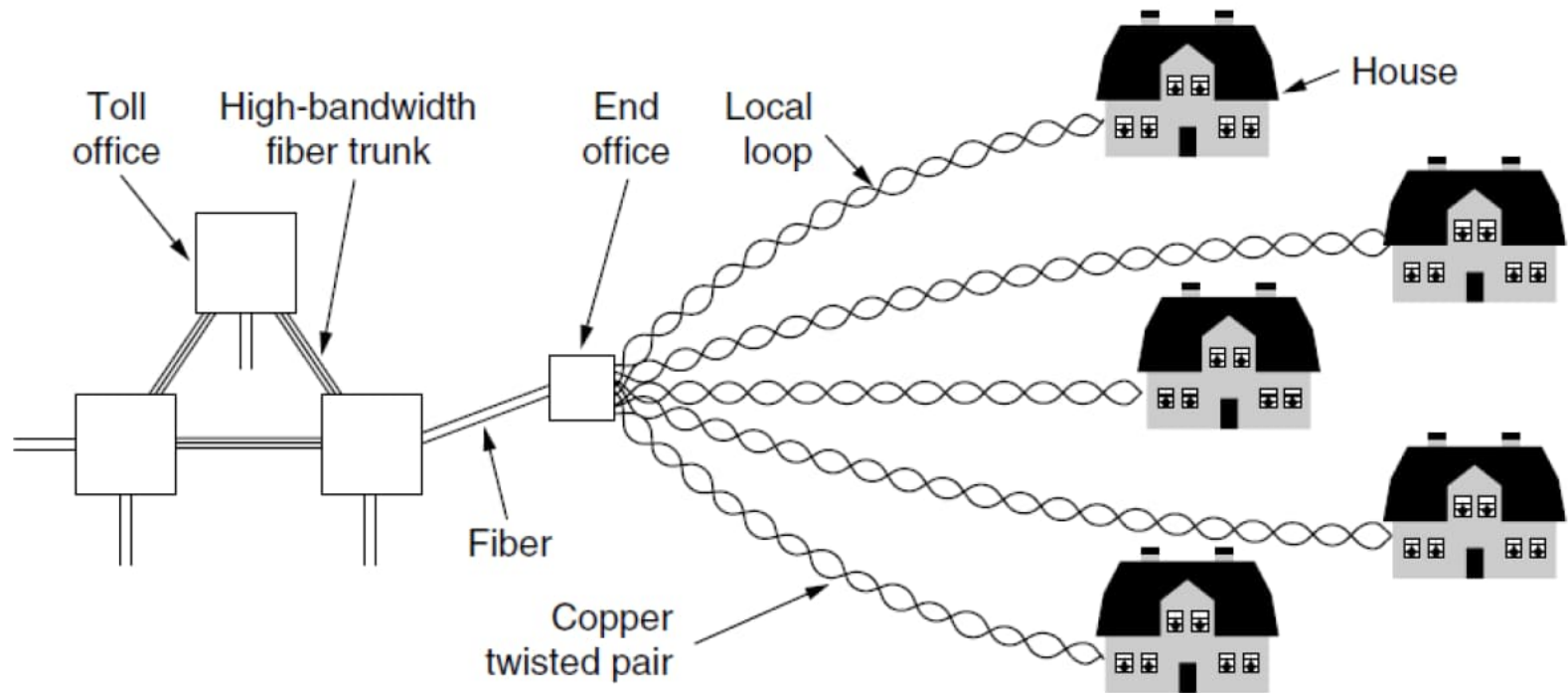


Figure 2-51. (b) The fixed telephone system.

Spectrum Allocation

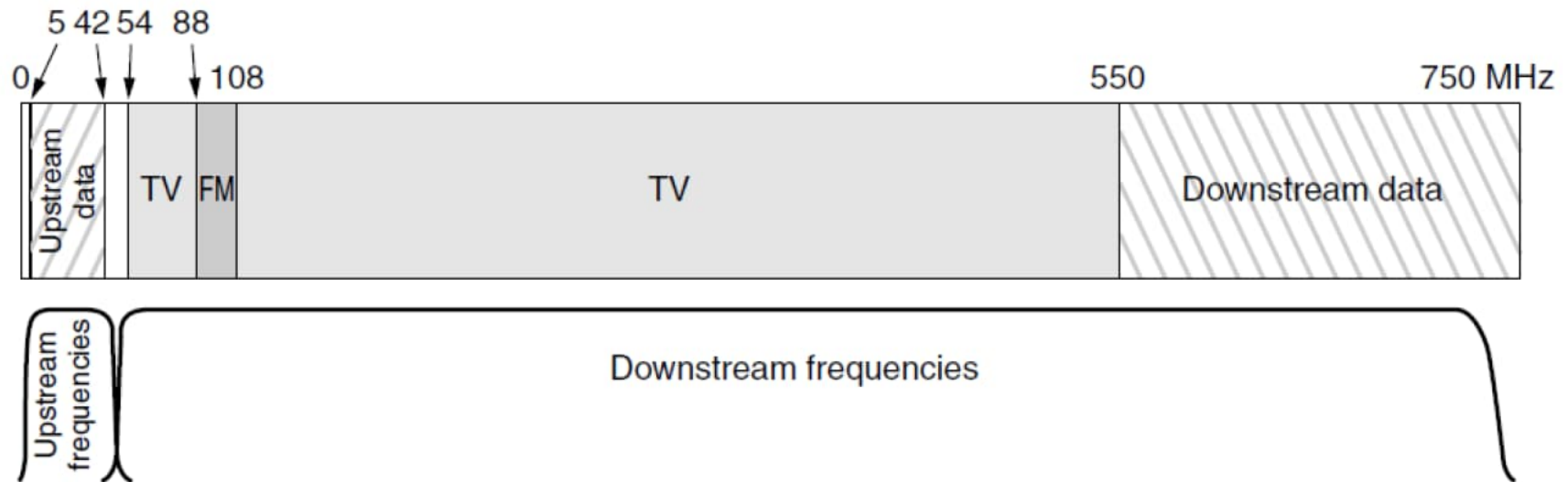


Figure 2-52. Frequency allocation in a typical cable TV system used for Internet access.

Cable Modems

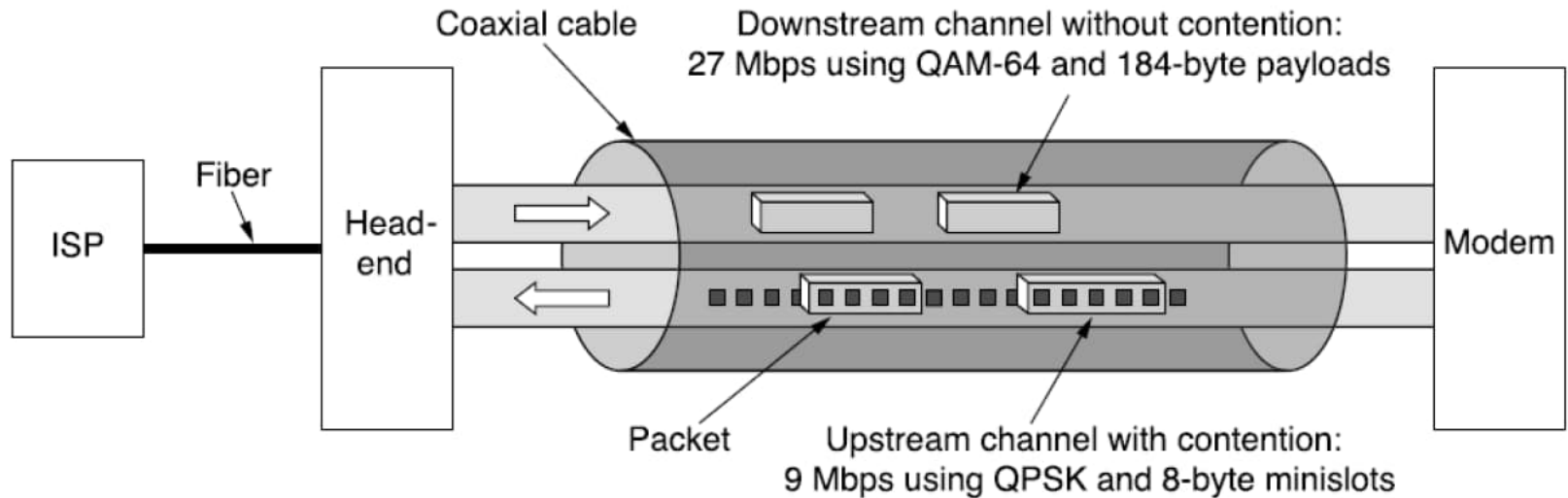


Figure 2-53. Typical details of the upstream and downstream channels in North America.

Summary

- Two **fundamental limits** on all channels
 - ✦ **Nyquist** limit, which deals with noiseless channels.
 - ✦ **Shannon** limit, which deals with noisy channels.
- **Transmission media** can be **guided** or **unguided**.
 - ✦ **Guided media** are **twisted pair**, **coaxial cable**, and **fiber optics**.
 - ✦ **Unguided media** include **terrestrial radio**, **microwaves**, **infrared**, **lasers through the air**, and **satellites**.
- **Digital modulation** methods
 - ✦ **baseband**: line codes.
 - ✦ **passband**: modulating the amplitude, frequency, and phase of a carrier.
- A key element in most **wide area** networks is the **telephone** system.
- An alternative system for network access is the **cable television** system.
- We will have a **class test** next Monday.